University Abdel-Hamid Mehri Constantine II

NTIC Faculty/DIFA 3° TI

Installation and Administration of Network Services.

IASR

Radouane NOUARA

subject objectives

allow students to acquire the necessary knowledge to :

- Understand the TCP/IP protocol.
- Install and configure Linux distributions.
- Configure a network of computers.
- **Install and configure network services**.
- **Administrate a network service**.
- Configure the firewalld service.

Evaluation

- **Evaluation :**
- Continuous evaluation (25%).
- Workshops (25%).
- Final Exam (50%).

- 3 Absences ➔ Exclusion du module.
- 5 Absences même justifiées ➔ Exclusion du module

To contact me

- I am available at any time on:

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Plan

- The OSI model.
- TCP/IP Protocol.
- Routing
- network configuration files.
- Tools for network administration.
- Network services in Linux.
- DHCP service.
- NFS server.
- SAMBA service.
- Management of the *firewalld* service.

Définitions

What is a network ?

Network definition

A network is a set of objects or people connected or maintained in connection.

- Un *réseau* est un ensemble ***d'objets*** ou de ***personnes*** connectées ou maintenues en liaison.
  - Social network.
  - Railway network.

Computer network

Definition of computer network

A computer network is **an interconnected system of two or more computing devices to transmit and share data**, **applications**, and other resources.

*Un réseau informatique désigne un ensemble de **machines** inter-connectées pour l'échange des flux de données.*

- ***Devices:*** *computer or any devices that can connect to the network.*
- ***Inter-connection support*** *: wire or wireless.*

Networks

Types of Networks:

- **PAN** : Personnal Area Network.
- **LAN** : Local Area Network.
- **MAN** : Metropolitan Area Network.
- **WAN :** Wide Area Network.

The Network history

- From the 60s.
- The increase need of data communication between enterprises (ex. between banks). (was blocked by **non-interoperability** of systems).
- Diversity of solutions (*Proprietary solutions*)
  - **NPL network in the UK**
  - **ARPANET in the USA**
  - **CYCLADES in France**
  - **SNA** -> ***IBM***.
  - **DSA** -> ***BULL***.
  - **DecNet** -> ***DEC***.

government-sponsored

Vendor-developed

- ***The existing solutions are incompatible and Lack of Interoperability*.**

The Network history

- **software interoperability :**

  Drivers and OS libraries used in network

- **Hardware interoperability :**

  Connectors, wires, electrical specificities

Obligation **to standardize** networks from different manufacturers to be able to interconnect them.

The Network history

Objective : Ensure the Network transparency.

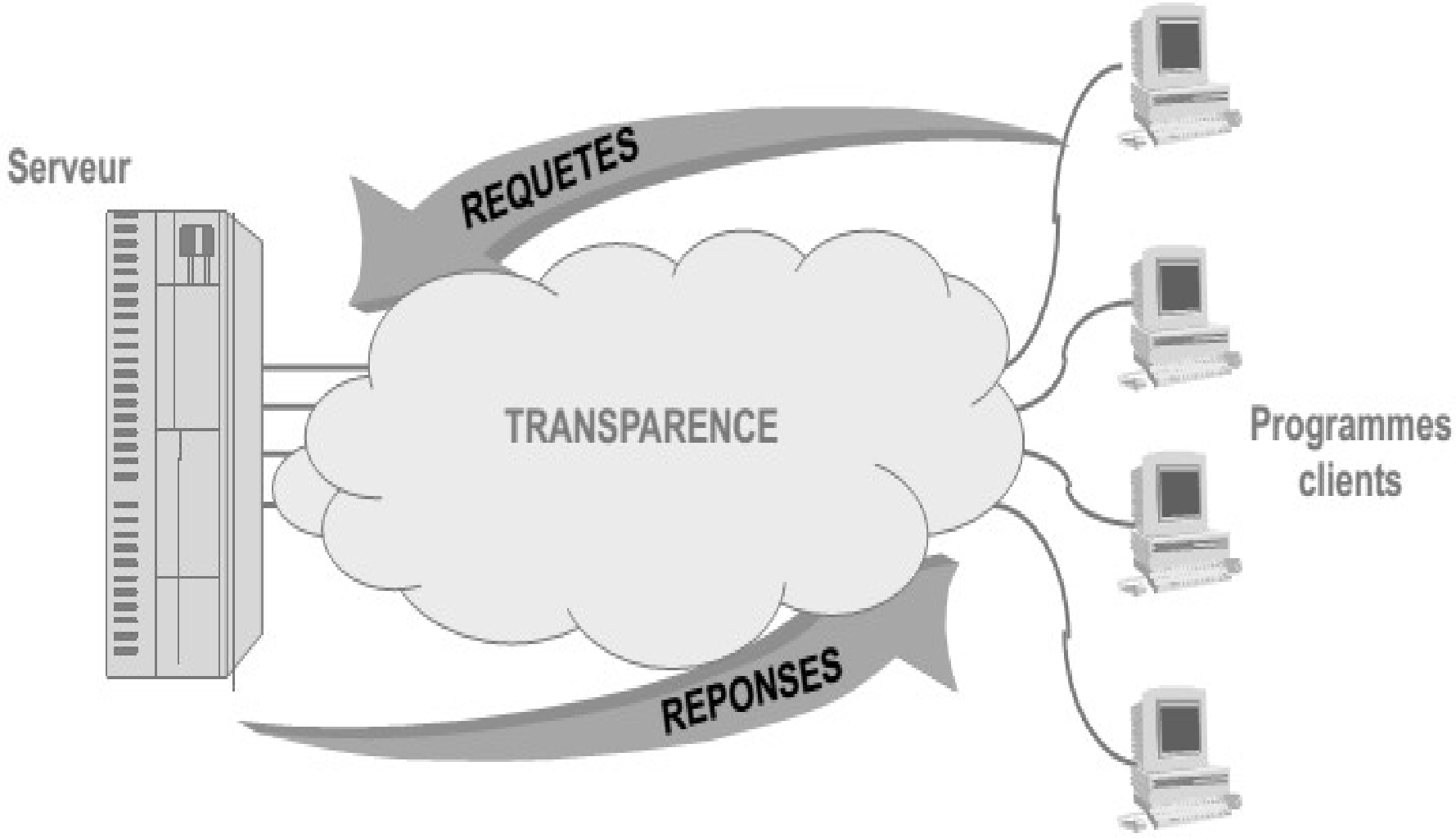
*The network must **ensure** the **transfer of data** between the transmitter and the receiver **without any modification or alteration**.*

Le réseau doit **assurer** le transfert des données entre l'émeteur et le récepteur sans aucune modification ou altération.

Ref: Réseaux & Telecom

Ex. The mail transfert.

the Network transparency



Ref: Réseaux & Telecom

OSI Model

- **ISO (***International Standardization Organization):* undertook the definition of a standardized communication architecture.
- Interconnect open systems.
- Architecture called reference model or **OSI** model (*Open System Interconnection)*.
- Connect equipment from different manufacturers in a **transparence** way.

OSI Model

- OSI model started in the late 1970s to support the emergence of the diverse computer networking methods.
- In the 1980s, the model became a working product of the OSI group.
- OSI had two major components: an abstract model of networking, called the Basic Reference Model or seven-layer model, and a set of specific protocols (developed jointly by the ISO and the ITU-T).

OSI Protocol stack

Requires the use :

- Compatible connection techniques

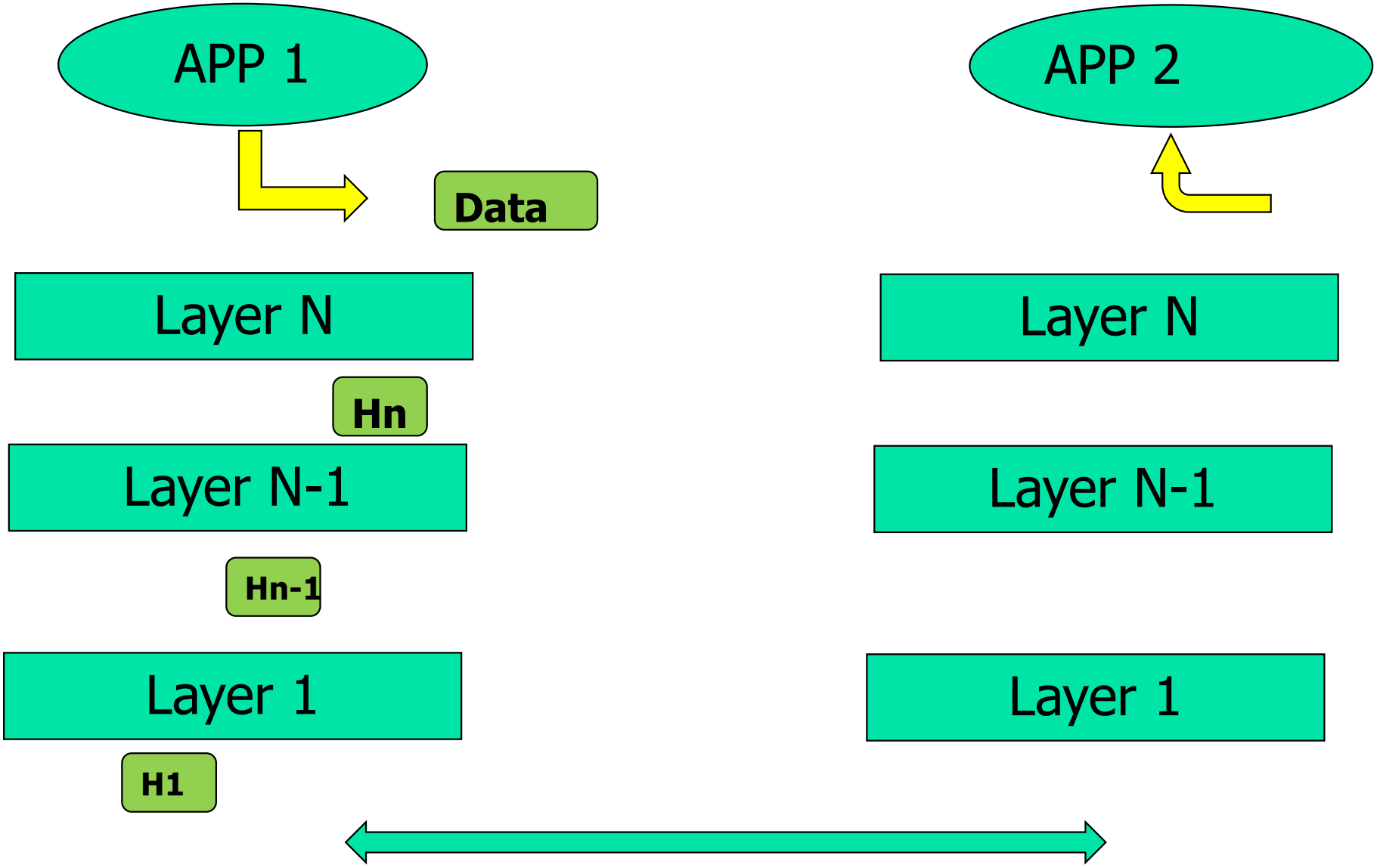
(connection raccordement, electrical level...).

- Identical exchange protocols.
- A semantics (the meaning) of information understandable by communication partners.

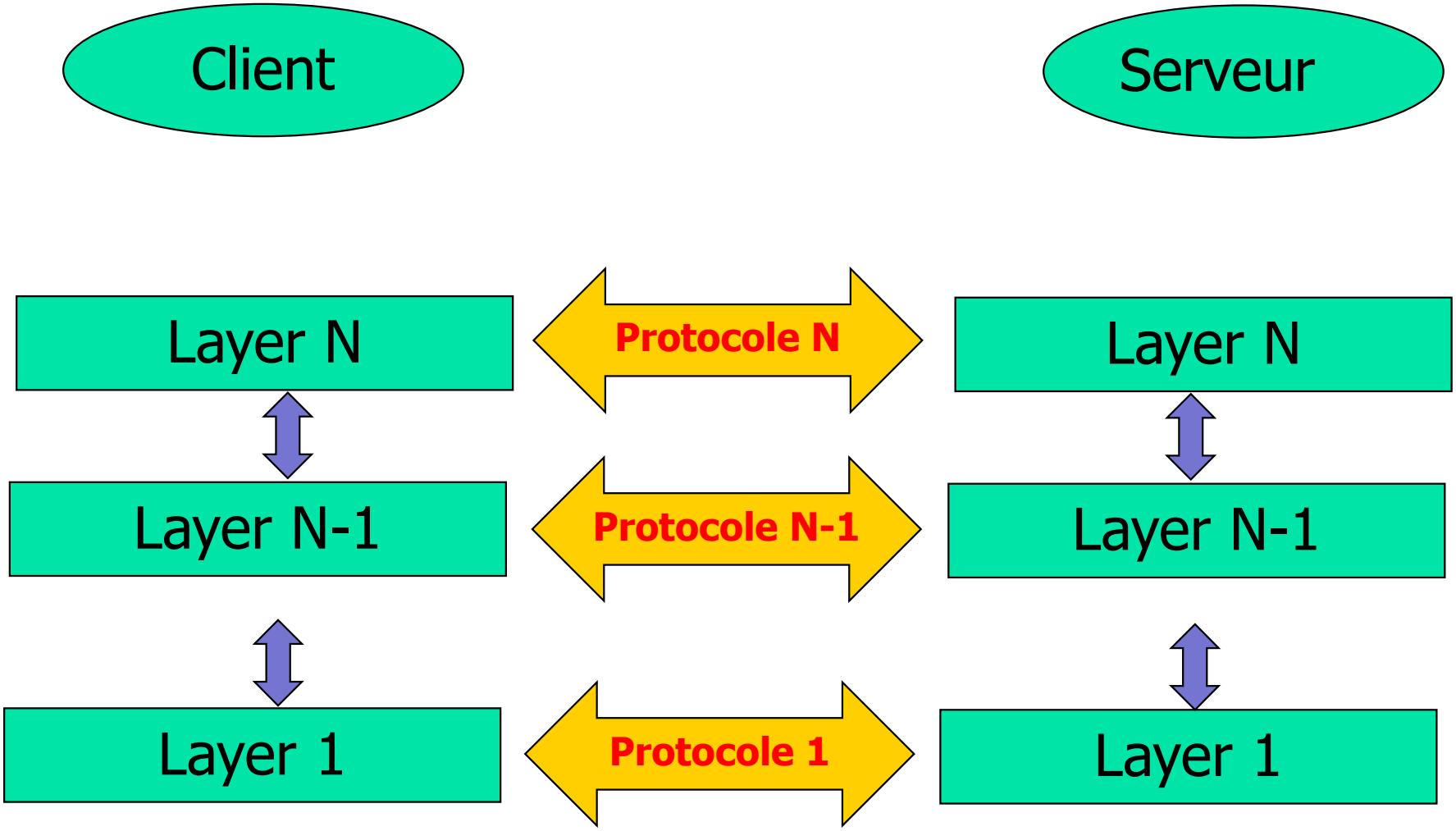
OSI Protocol stack

- To overcome this compexity
- Avoid a complexe description of the solution.
- The solution consists to split the system in a set functionnal entities called Layers.
- Each layer takes care of part of the communication task.

OSI Protocol stack



OSI Protocol stack



Concept of protocol

- A protocol is a set of conventions pre-established to achieve a reliable exchange of data between two entities.
- the delimitation of the data blocks exchanged.
- Checking the integrity of the data received.
- The organization and control of the exchange.

OSI model Stack

- The OSI model is composed of 7 Layers.

Application Layer
Presentation Layer
Session Layer
Transport Layer
Network Layer
Data Link Layer
Physical Layer

Physical Layer

- Ensures bit transfer on the physical channel (medium).
- Defines the supports and means of accessing the channel.
- Establishment, maintenance and release of the channel.

Data Link Layer

- Provides a data block (frame) transfer service between two adjacent systems.
- ensures the control, establishment, maintenance and release of the **logical link** between entities.
- detects and corrects errors inherent in physical media.

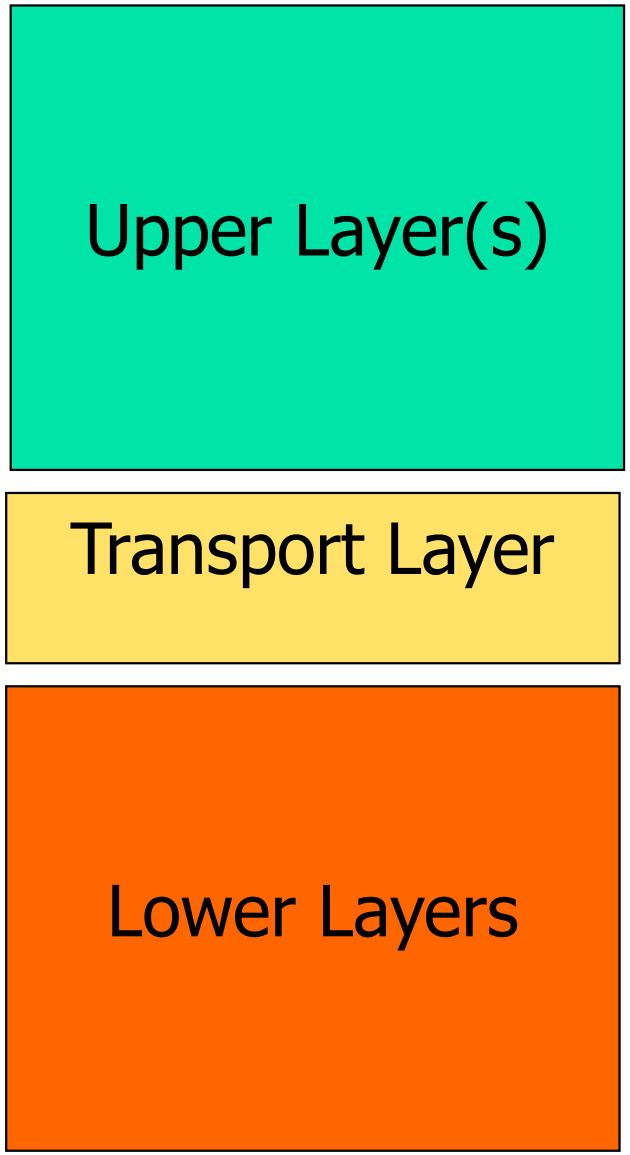
Network Layer

- Ensures the routing of data (packets) through the different nodes of a subnetwork (ROUTING).

- provides the means to ensure :
  - call routing,
  - The Packets routing,
  - Congestion control,
  - Adapting the size of the **data blocks** to the capacities of the used physical subnetwork.

Transport Layer

- It ensures control of the end-to-end transfer of data (**messages**) between the two end systems.
- it ensure reliable transfer to the upper layers regardless of the quality of the transport subnetwork used.



Session Layer

- Manages the exchange of data (transaction) between remote applications

- Allows the synchronization of exchanges and the definition of recovery points.

Presentation Layer

- Constitutes the interface between the layers which ensure the exchange of data and the one which manipulates them.
- Layer not involved in the information transfer mechanisms ensures the formatting of the data.
- In addition to other special functions, such as data compression...

Application Layer

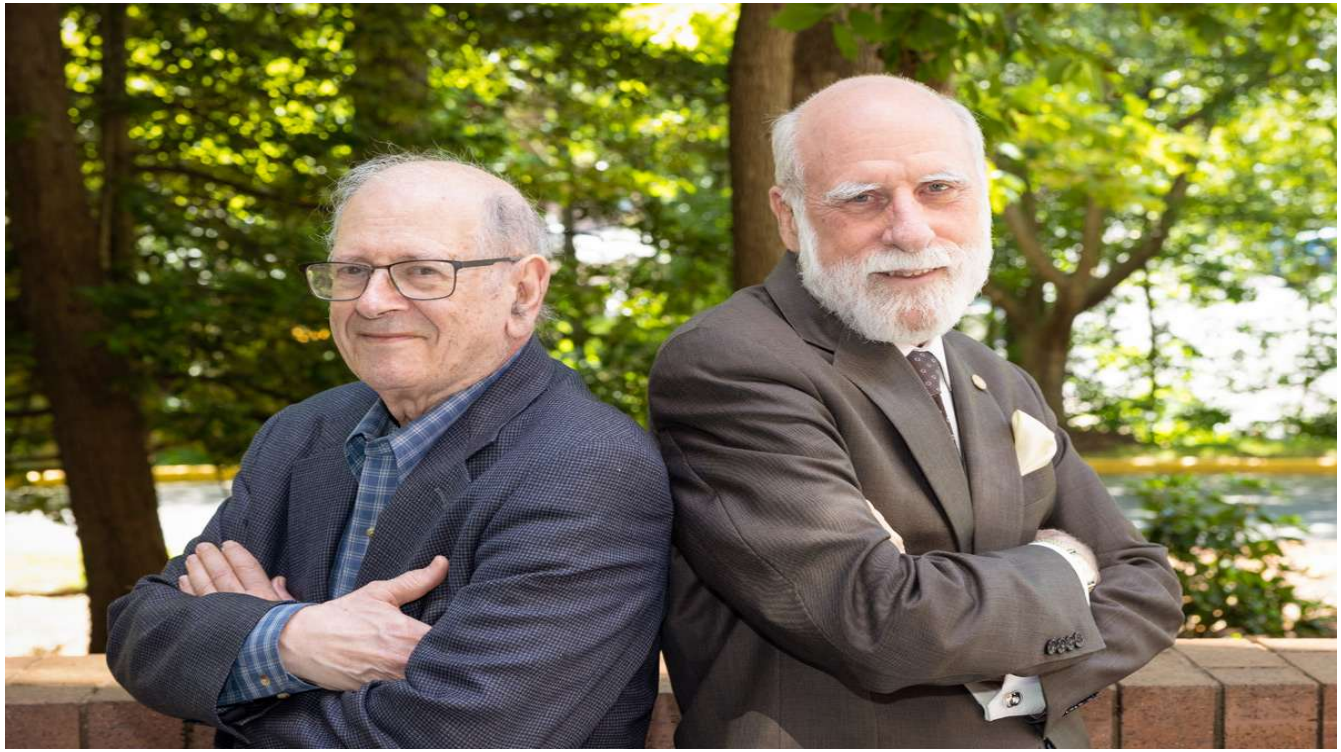
- The used applications
- Ex. Browser, mail app, …..

TCP/IP Stack

- Transmission Control Protocol/Internet Protocol.
- Developed in the 1970s by DARPA for the needs of communication and interworking of applications between the US army computer systems (**DoD)**.
- Guarantee high resistance to network node failure.

TCP/IP Stack

- Vinton Gray Cerf and Robert E. Kahn propose the TCP/IP protocol suite to improve communication.
- In 1981 the protocol were documented in RFCs to became the official documentation of the internet.



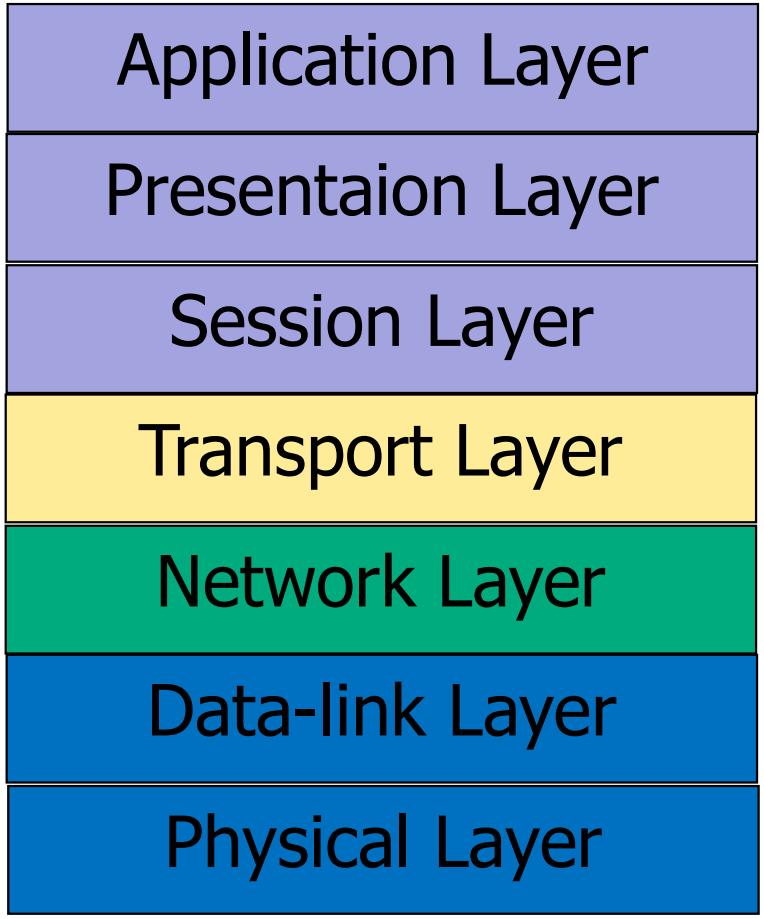
TCP/IP Stack

- A set of protocols to solve interconnection problems in heterogeneous environments.
- describes a logical network (Called IP network) over the physical networks.
- The integration of TCP/IP to UNIX BSD 4 (1983) made it a standard in Unix and then Linux communauties.

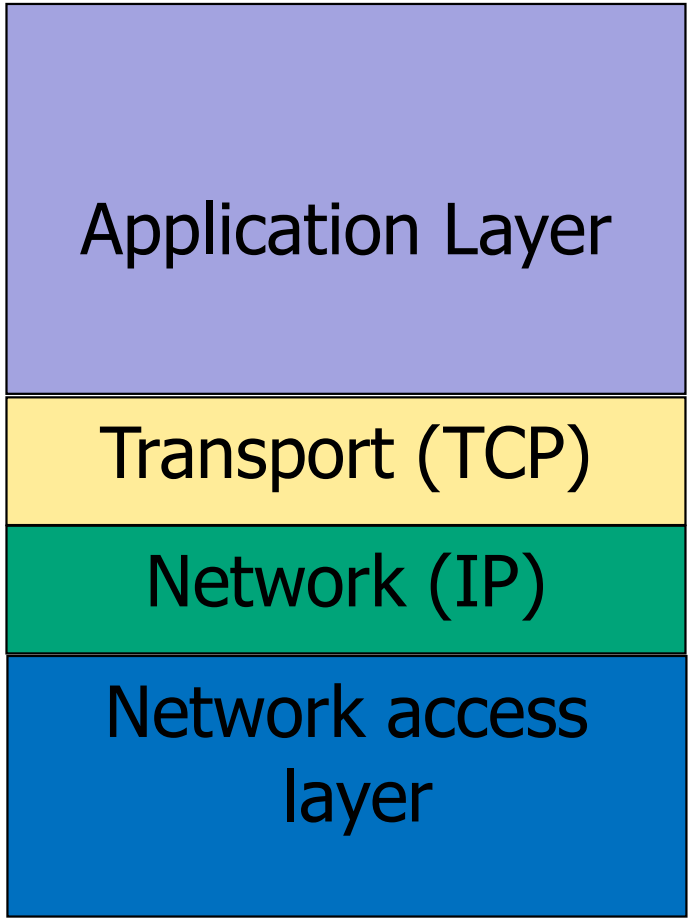
TCP/IP Stack

- **Layered architecture**: Organizes network communication into **4 layers**, with each layer performing specific tasks.
- **Reliability**: Ensures data integrity through mechanisms like error detection, correction, and retransmission.
- **Scalability**: The protocol is designed to support networks of all sizes.
- **Hardware independence**: TCP/IP can work across different types of hardware and network technologies.

OSI vs TCP/IP



OSI

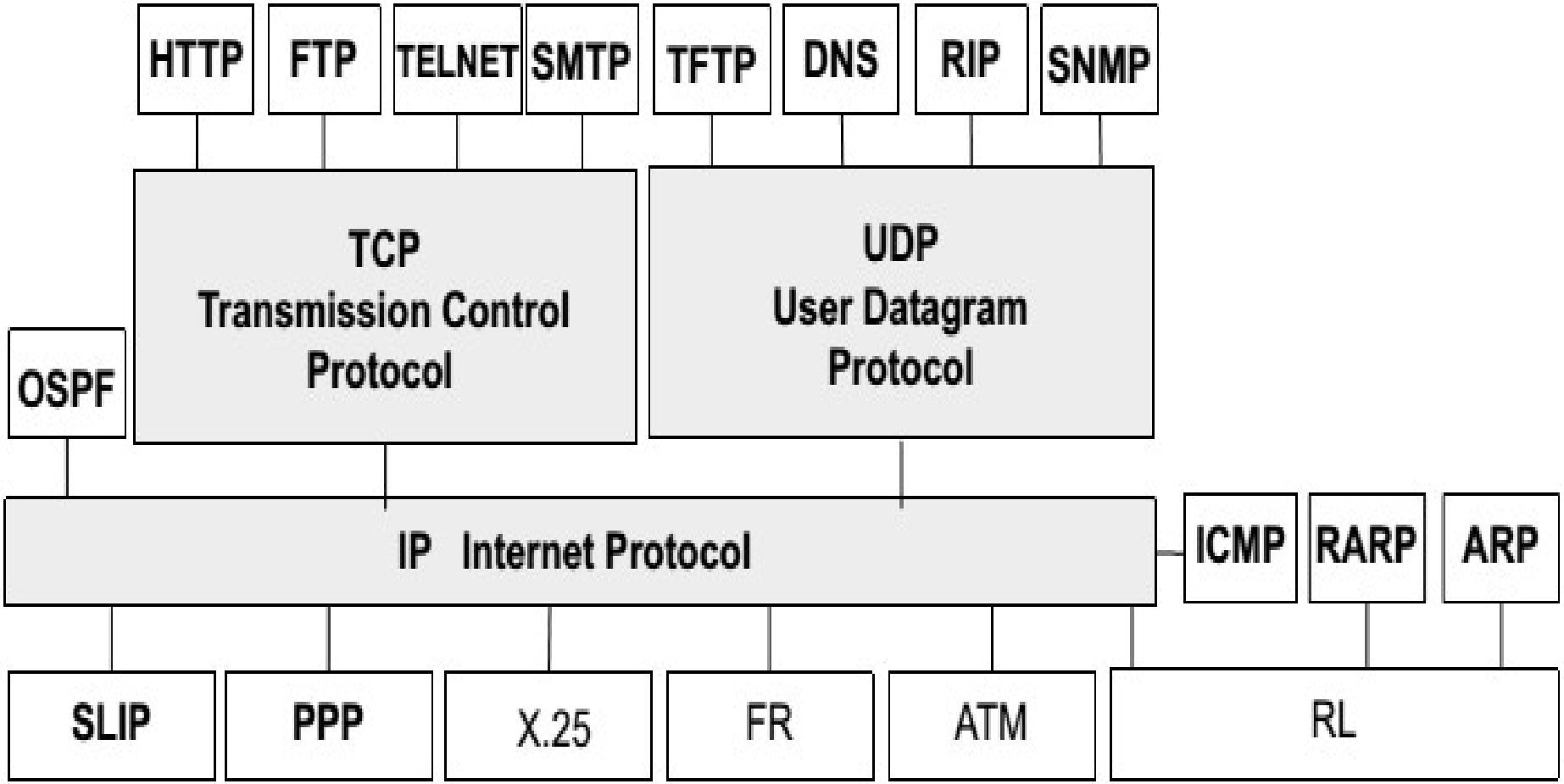


TCP/IP

TCP/IP Stack

- The transport layer provides two types of service :
- **TCP:** Service in connected mode (similar to TP4 of ISO).
- **UDP:** (*User Datagram Protocol*) A light-weight transport service offering only a *best effort* (datagramme).

TCP/IP Stack



Ref: Réseaux & Telecom

Network Access Layer

- PPP : Point to Point protocol
- SLIP:
- X.25 is dead.
- FR: Frame Relay
- ATM : Asynchronous transfer mode
- RL :

TCP/IP Stack

- **ARP :** Correspondance **IP** to **MAC**.
- **RARP**: Correspondance **MAC** to **IP**
- **DNS**: correspondance domaine name to IP.
- **ICMP:** enables control or error messages to be sent to other machines or gateways.

  (ex Destination Unreachable)

TCP/IP Stack

- **OSPF :** (*Open Shortest Path First):* est un protocole de routage interne IP de type « à état de liens ».
- ***SLIP : (Serial Line Internet Protocol***) : un protocole de liaison simple ne fournissant aucun contrôle d'erreur ou d'adresse.
- **PPP: (Point-to-Point Protocol):** de transmission pour l'internet, basé sur HDLC.

TCP/IP: Data Encapsulation

- Transporting data from one layer to a data unit on the lower layer.
- The lower layer adds a header, used by the remote peer entity to extract the data.
-

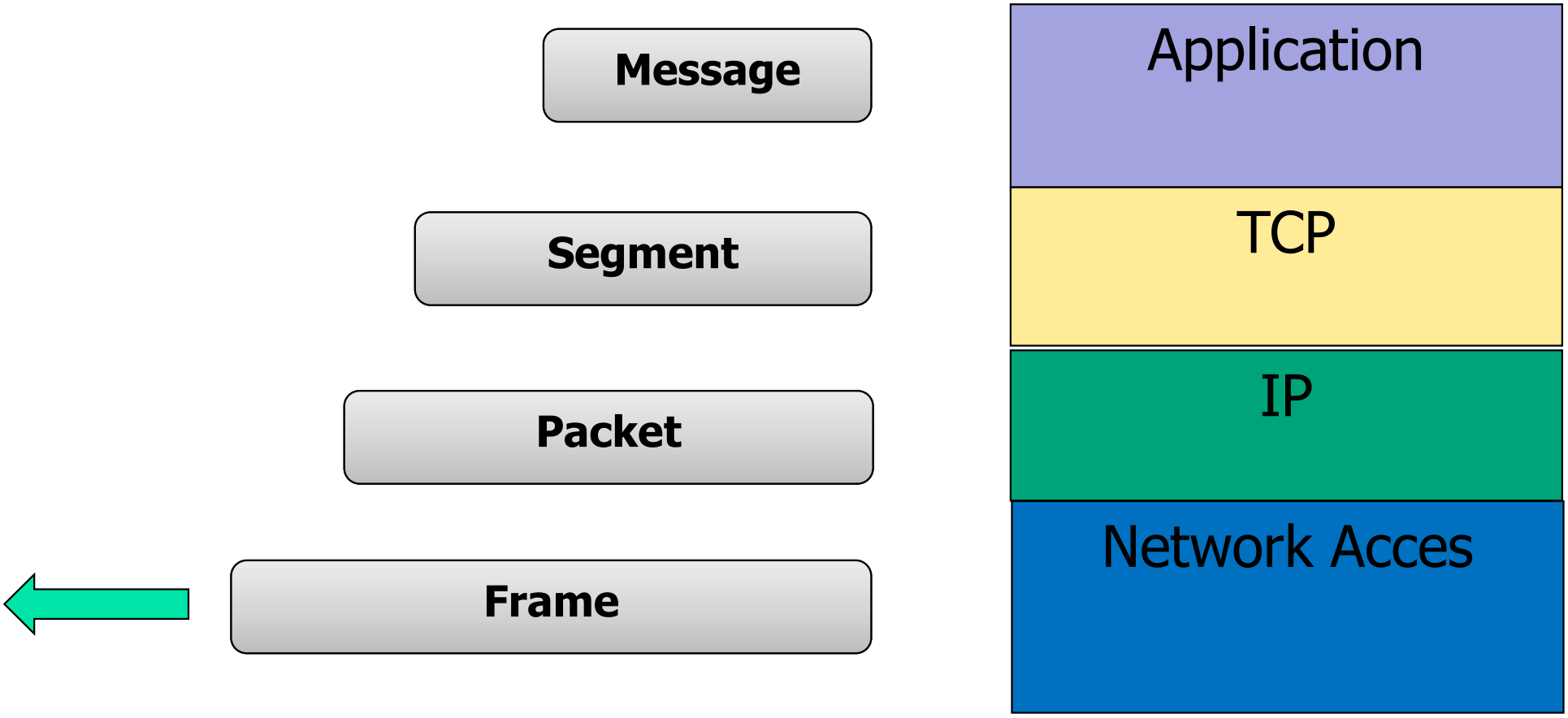
Network Access Layer Protocols

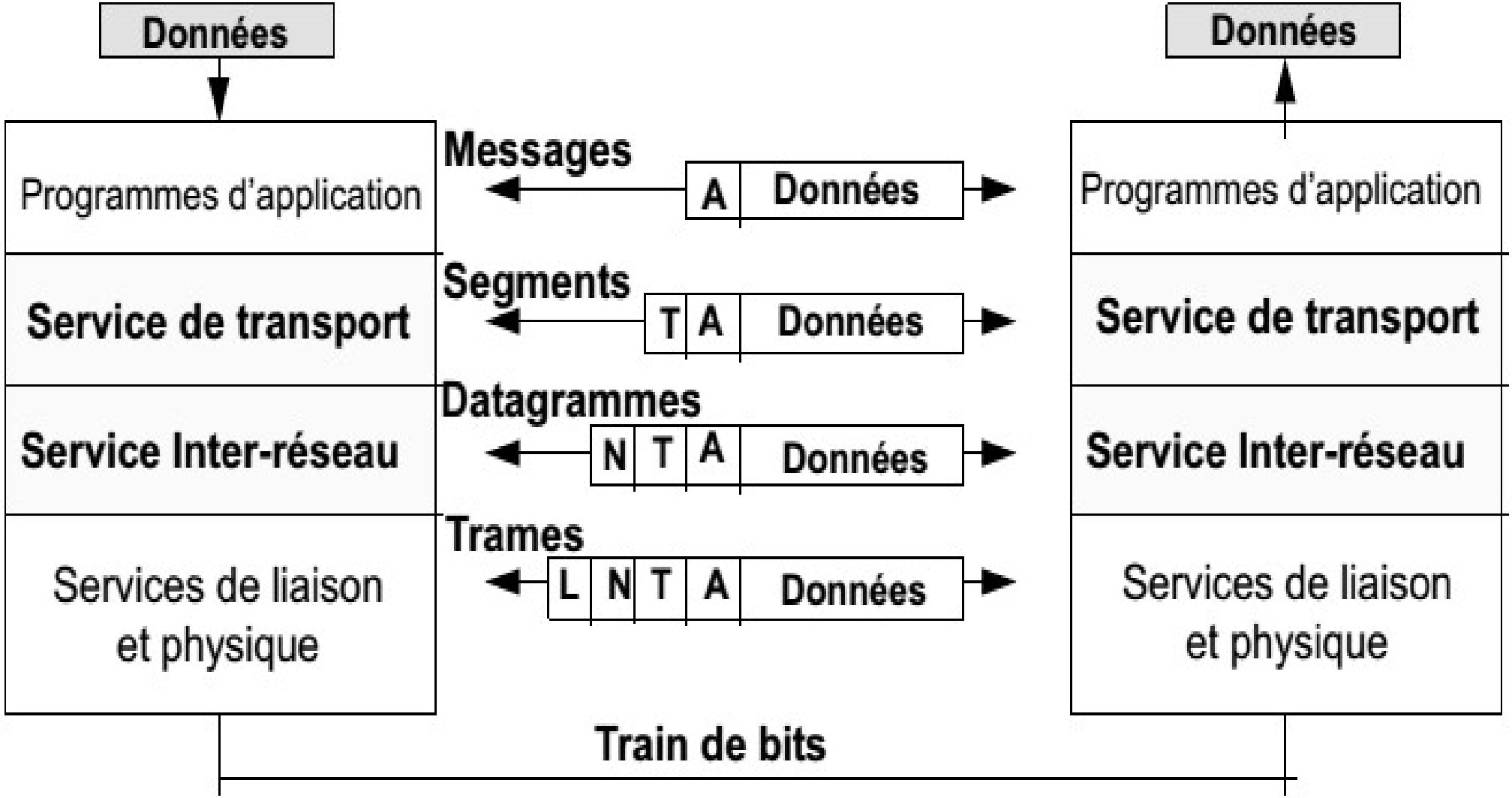
- **Ethernet:** For wired LANs that defines.
- **Wi-Fi (IEEE 802.11)**: For wireless communication.
- **Point-to-Point Protocol (PPP)**: A protocol often used for direct connections between two network nodes, typically on wide area networks (WANs).

Network Address Layer.

- **Ethernet**: For wired LANs that defines how devices communicate on the network using MAC addresses and frames.
- **Wi-Fi (IEEE 802.11):** The standard for wireless communication.
- **Point-to-Point Protocol (PPP):** A protocol used for direct connections between two network nodes.

TCP/IP: Data Layers





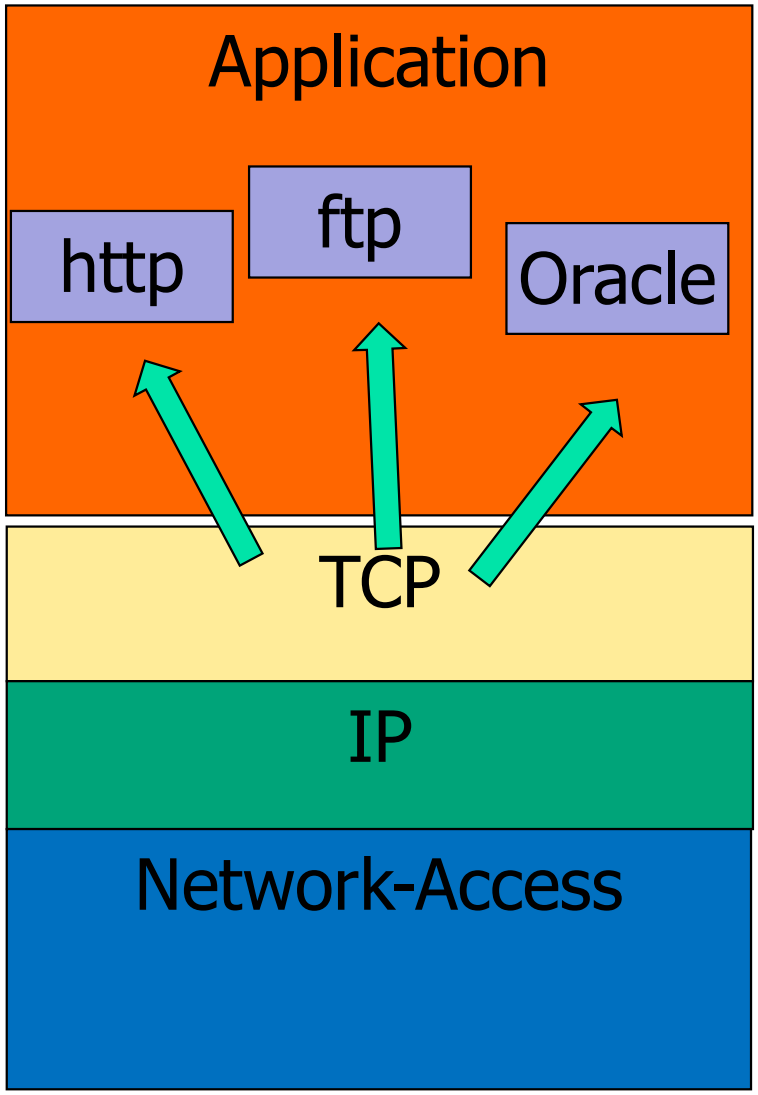
Ref: Réseau & Telecom

TCP/IP: Addresses of Applications.

- Several applications can run on the same computer in a network.
- To differentiate them, we use the notion of ***port***.
- Each application will have a unique number that identifies it, this number is called a Port.

TCP/IP: Addresses of Applications.

How to identify the application receiving the data?



Data

TCP/IP: Ports

- A number on 16 bits.
- *IANA (Internet Assigned Numbers Authority) is responsible for port assignment.*
- ***ports** from 0 to 1023 are reserved (well known).*

- Ports from 1024 to 49151 are **Registered Ports**.

- Ports from 49152 to 65535 D**ynamic and/or Private Ports**.

TCP/IP: Ports

Reserved Ports		Registered Ports	
Port	Application	Port	Application
20 et 21	FTP		
22	SSH		
23	Telnet	1433, 1434	SQL Server
25	SMTP	1521	Oracle
53	DNS	3128	Squid
80	HTTP	3306	MySQL
110	POP	3390	Xbox
143	IMAP3		

TCP/IP: Ports

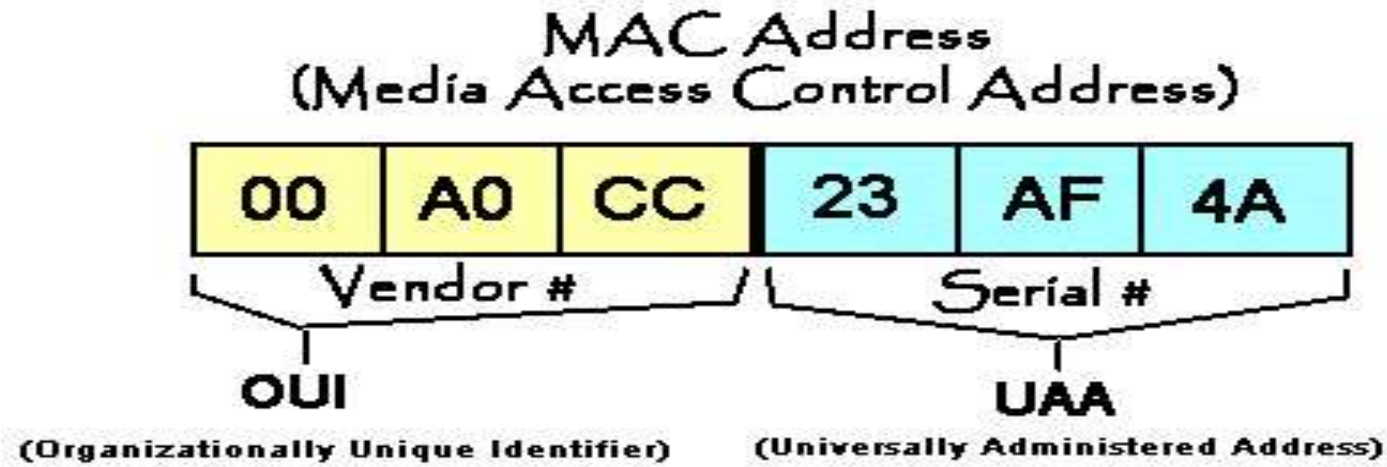
- On Linux, to list running apps with their ports : *ss -tulpn*

```
[root@pxeserv ~]# ss -tulpn
Netid  State      Recv-Q Send-Q     Local Address:Port       Peer Address:Port
udp    UNCONN     0      0                     *:69                    *:*
 users:(("xinetd",1052,5))
tcp    LISTEN     0      32                    *:21                    *:*
 users:(("vsftpd",1066,3))
tcp    LISTEN     0      128                  ::22                    ::*
 users:(("sshd",1041,4))
tcp    LISTEN     0      128                   *:22                    *:*
 users:(("sshd",1041,3))
tcp    LISTEN     0      100                 ::1:25                    ::*
 users:(("master",1320,13))
tcp    LISTEN     0      100          127.0.0.1:25                    *:*
 users:(("master",1320,12))
tcp    LISTEN     0      50         192.168.56.2:3306                 *:*
 users:(("mysqld",1211,10))
tcp    LISTEN     0      128                  ::80                    ::*
 users:(("httpd",1333,4),("httpd",1336,4),("httpd",1337,4),("httpd",1338,4),("ht
tpd",1339,4),("httpd",1340,4),("httpd",1341,4),("httpd",1342,4),("httpd",1343,4)
)
[root@pxeserv ~]# _
```

Physical Address MAC

- A MAC address (Media Access Control) is a 6 bytes (12-digit hexadecimal number) assigned to each device connected to the network.
- Primarily specified as a unique identifier during device manufacturing.
- the MAC address is often found on a device's network interface card (NIC).
- Unique in the world.

Physical Address (MAC).



OUI :Organizationally Unique Identifier that uniquely identifies vendor, manufacturer or other organization..

UAA: A sequential number attributed by the manufacturer.

Numéro constructeur (en hexadécimal)	Constructeur	Numéro constructeur (en hexadécimal)	Constructeur
00-00-0C	Cisco	08-00-02	3Com-Bridge
00-00-0F	NexT	08-00-05	Symbolics
00-00-10	Sytek	08-00-06	Siemens Nixdorf
00-00-1D	Cabletron	08-00-07	Apple
00-00-2A	TRW	08-00-09	HP
00-00-5E	IANA	08-00-0A	Nestar Systems
00-00-65	Network General	08-00-0B	Unisys
00-00-6B	MIPS	08-00-10	AT&T
00-00-77	MIPS	08-00-11	Tektronics
00-00-81	Synoptics	08-00-14	Excelan
00-00-89	Cayman Systems	08-00-20	Sun
00-00-93	Proteon	08-00-2B	DEC
00-00-A2	Wellfleet	08-00-38	Bull
00-00-A7	NCD	08-00-39	Spider
00-00-A9	Network Systems	08-00-46	Sony
00-00-AA	Xerox	08-00-4E	BICC
00-00-C0	Western Digital	08-00-5A	IBM
00-00-C9	Emulex	08-00-69	Silicon Graphics
00-AA-00	Intel	08-00-6E	Excelan
00-DD-00	Ungermann-Bass	08-00-7C	Vitalink
00-DD-01	Ungermann-Bass	08-00-90	Retix

MAC

- in Linux OS the CMD ip a gives NIC with their MAC adresses.

```
[root@pxeserv ~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP ql
en 1000
    link/ether 08:00:27:0a:14:87 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.50/24 brd 192.168.1.255 scope global eth2
    inet6 fe80::a00:27ff:fe0a:1487/64 scope link
       valid_lft forever preferred_lft forever
3: eth3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP ql
en 1000
    link/ether 08:00:27:e1:32:d0 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.2/24 brd 192.168.56.255 scope global eth3
    inet6 fe80::a00:27ff:fee1:32d0/64 scope link
       valid_lft forever preferred_lft forever
[root@pxeserv ~]# _
```

IP Addressing

- IP is a connectionless protocol (no need to establish a connection)
- Every datagramme is routed independently.
- Datagrams can be mixed (not in the order of sending), duplicated or altered.
- IP is a *best-effort delivery (*network does not provide any guarantee that data is delivered or that delivery meets any quality of service).

IP Addressing

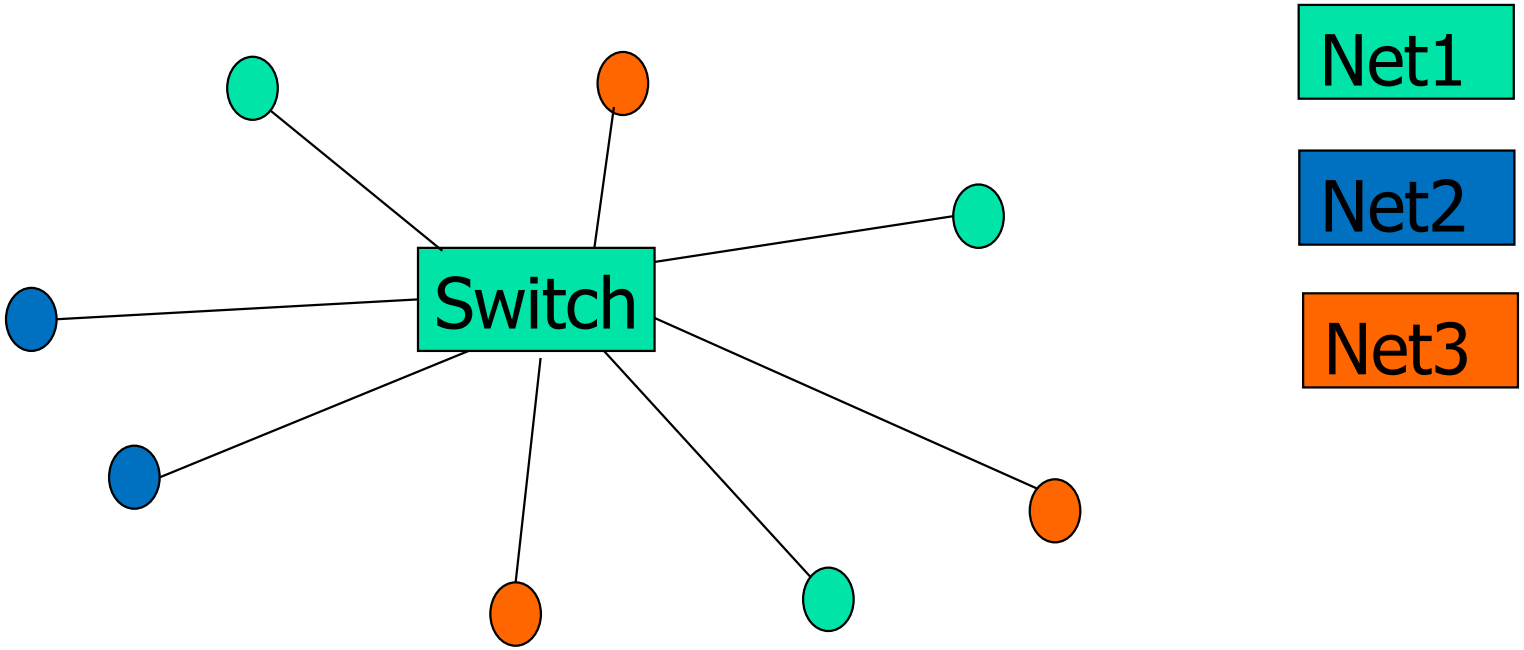
IP Protocol dont garantee :

- Data corruption.
- Order of packets arrival.
- Loss or destruction of packages.
- Packets duplication.

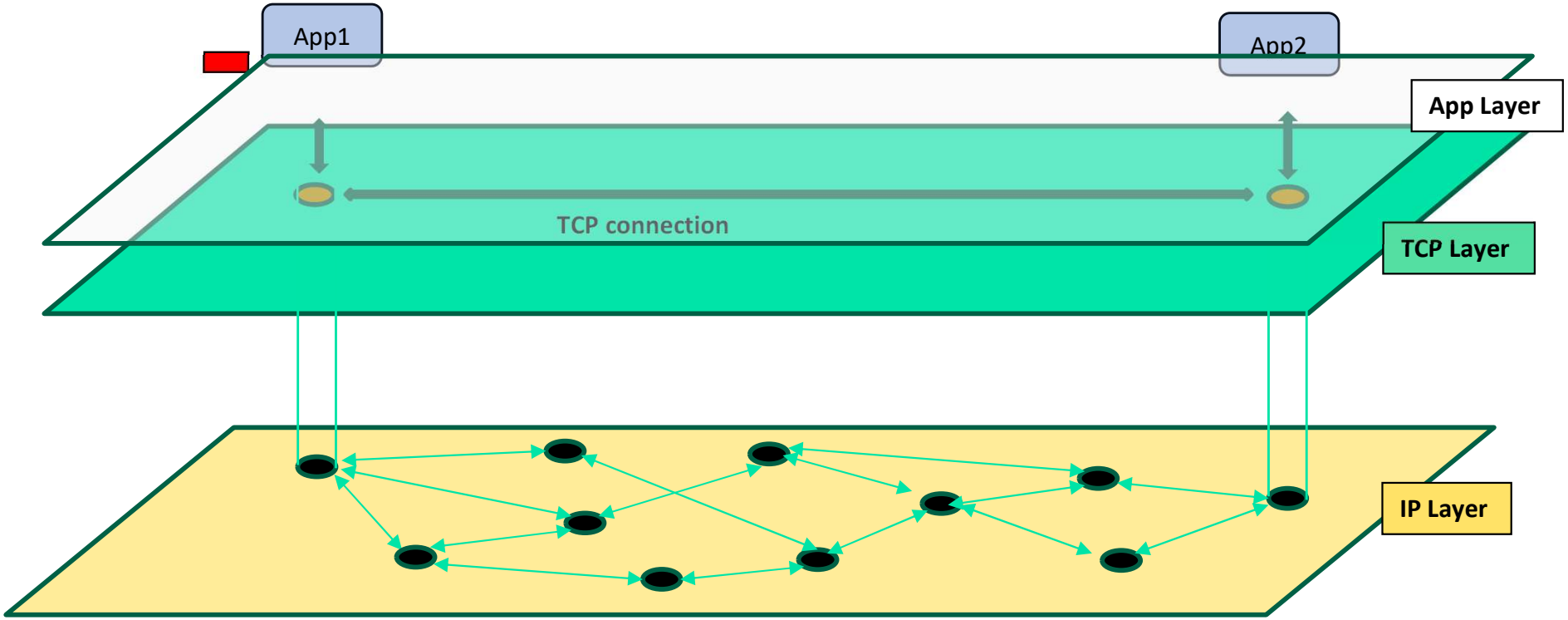
These tasks are performed by the upper layers.

IP Addressing

- TCP/IP offers a logical vision of the network, making it independent of the underlying hardware technologies.



IP Addressing



IP Addressing

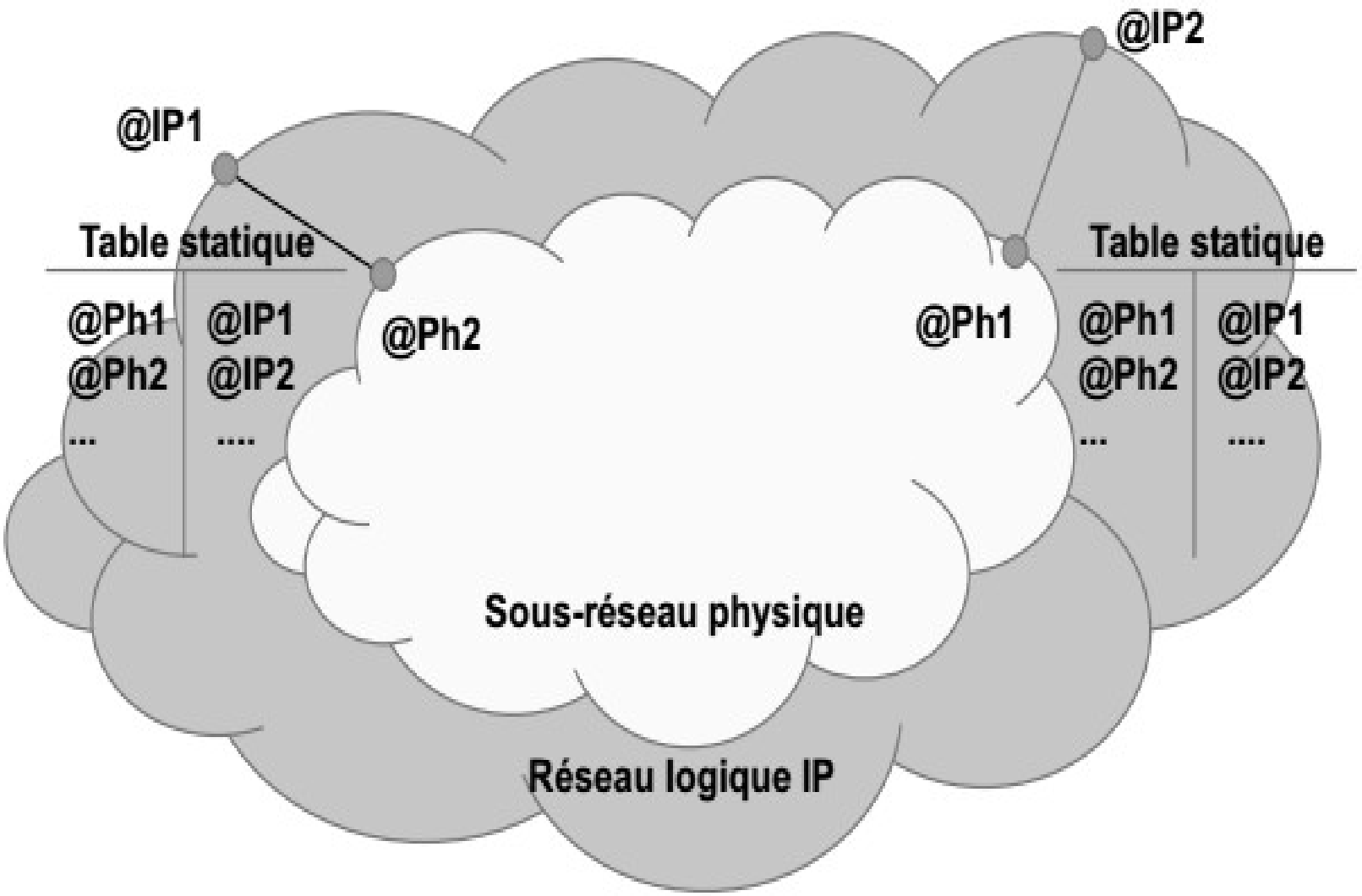
- Each ***computer*** is assigned a logical address called IP, that uniquely identifies it.

- The IP address is made up of 4 bytes (32bits) called **IPv4 address**.

IP Addressing

- Independence between logical addressing (in the network layer) and physical addressing (in the physical layer) (**MAC vs IP**).
- The need for a matching mechanism to translate an IP address to its corresponding MAC.
-

IP Addressing



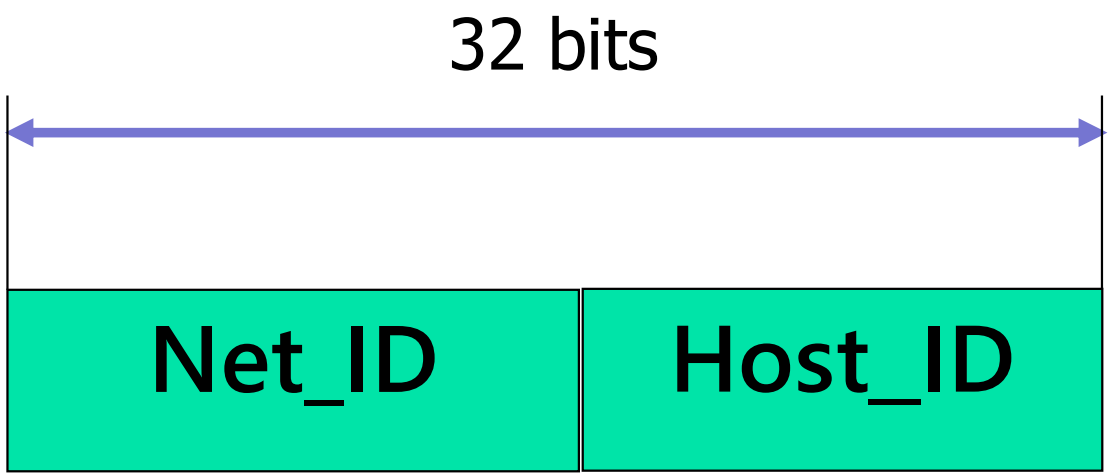
IP Addressing

- An IP address is noted as 4 decimal number (between 0 and 255) se.
- **Example**
- **193.27.45.33** in decimal correspond to:

- **11000001 00011011 00101101 00100001**

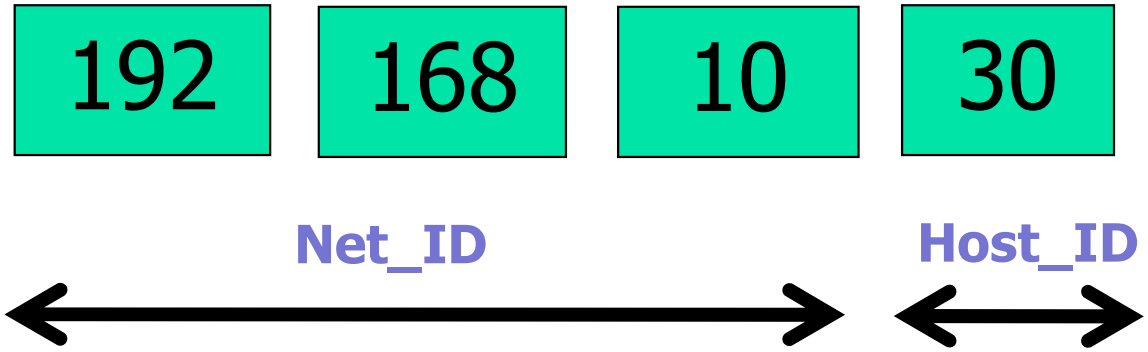
IP Classes

- An IP address is composed of two parts.
- The network Identifier **Net_ID**.
- The Host Identifier in the network **Host_ID**.



IP Classes

- IP address does not identify where **Net_ID** ends and **Host_ID** begins.



- To identify the **Net_ID** and where the **Host_ID** begins, a network mask is added.

Network Mask

- A Netmask is an IP address (32 bits).
- With bits set to 1 for the **Net_ID**
- And bits set to 0 for the **Host_ID**.

IP	192	168	10	30
Mask	1111111	1111111	1111111	00000000
	255	**255**	**255**	**0**

Net_ID ⟷ **Host_ID** ⟷

Network Mask

- IP = 192.168.10.30
- Mask =255.255.255.0
- The mask can be noted as a number of bits used for **Net_ID**.

192.168.10.30/24

Classless Inter-Domain Routing (CIDR)

Classe of IP adresses

- In order to adapt the size of both parties (Net_Id and Host_ID) to the company network.
- For a company with a large number of machines, the Net_ID would be short and the Host_ID long, and vice versa.
- For a small company, a long Net_ID and a short Host_ID.

Classe of IP adresses

- 5 classes of IP addresses are defined by **ICANN** (*Internet Corporation for Assigned Names and Numbers*).
- Classes from A to E.

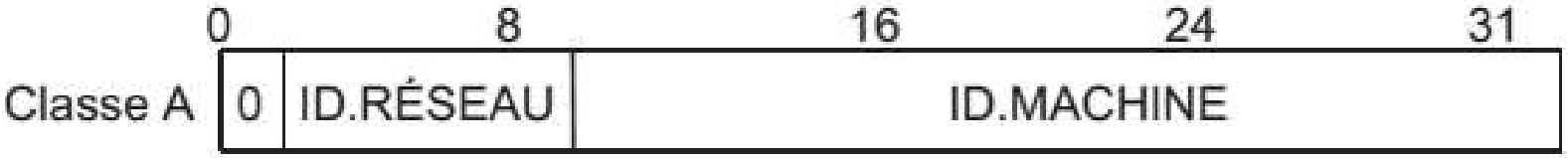
Classe of IP adresses

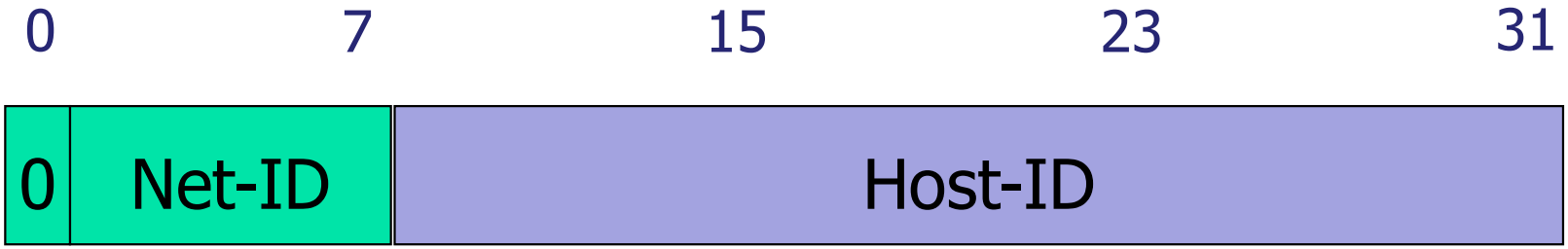
- A-class : for large-scale networks.

adresses from **1.0.0.1** to **126.255.255.254**

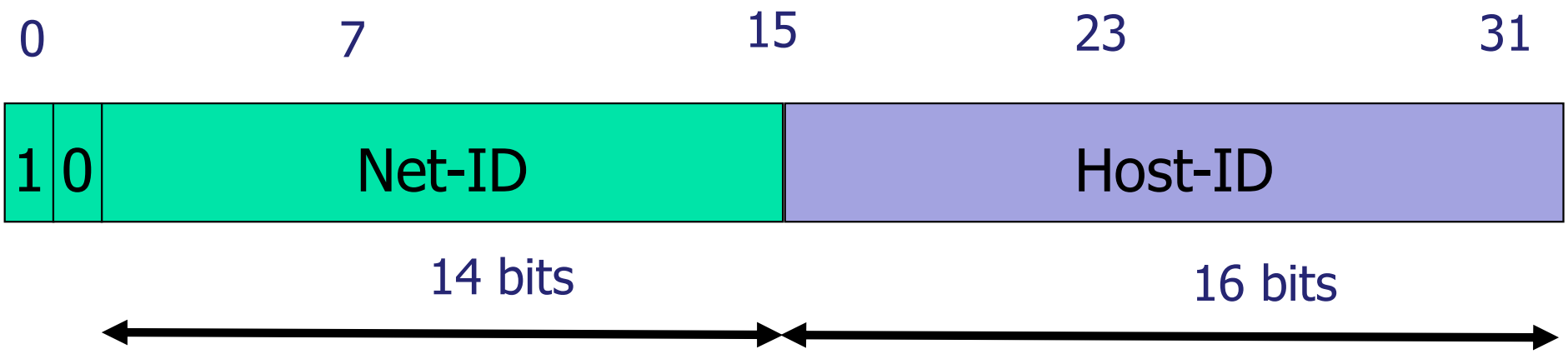
- 7 bits for **Net_ID** (126 networks)
- 24 bits for **Host_ID** (2^{24}-2 =16777214 Machines).
- Values between **0** and **127** are reserved

Classe of IP adresses





Classe of IP adresses



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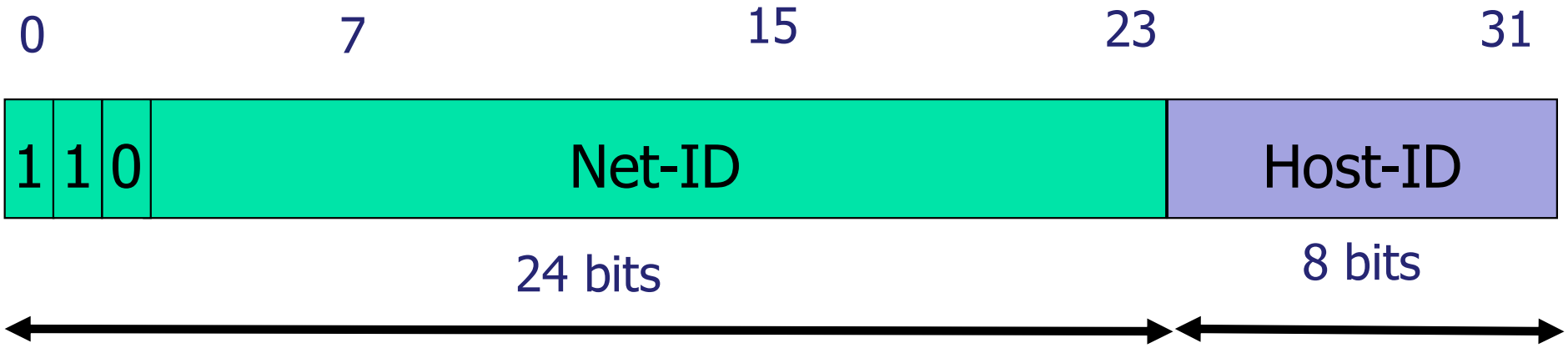
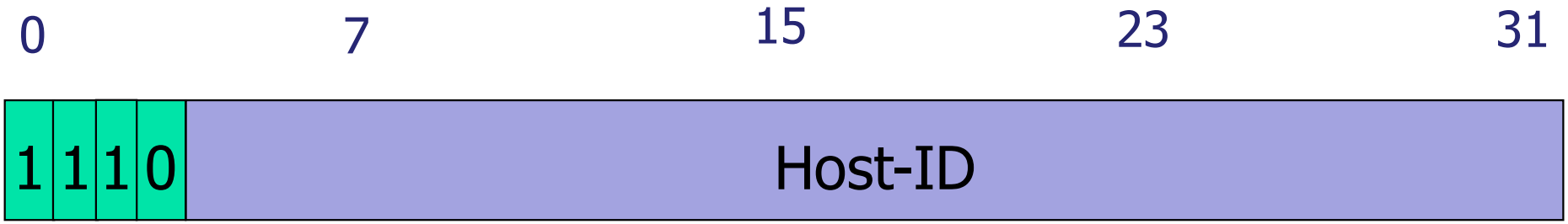


Classe of IP adresses



Classe of IP adresses

Sample A-Class Networks /8	
9.0.0.0	IBM Corporation
11.0.0.0	DoD Intel. Information Systems
12.0.0.0	AT&T ITS
17.0.0.0	Apple Computer, Inc.
18.0.0.0	Massachusetts Institute of Technology
19.0.0.0	Ford Motor Company
34.0.0.0	Halliburton Company
40.0.0.0	Eli Lilly and Company
43.0.0.0	Japan Inet
36.0.0.0	Stanford University

Classe of IP adresses

Classe B	1	0	ID.RÉSEAU	ID.MACHINE

- B-Class :for medium network.
- **128.0.0.1 to 191.255.255.254**
- 14 bits for **Net_ID** (16384 networks)
- 16 bits for **Host_ID** (2^{16}-2 =65533 Machines/Network)

Classe of IP adresses

Sample B-Class Networks	
128.004.0.0	University of Delaware
128.008.0.0	University of Maryland
128.088.0.0	Hewlett-Packard Company
128.094.0.0	Lucent Technologies
128.107.0.0 and (108)	Cisco Systems, Inc.
129.073.0.0	Siemens Corporate Research, Inc.
136.163.0.0	Saab Scania Aircraft
168.116.0.0	McGraw-Hill, Inc.
168.127.0.0	Fujitsu Network Switching
168.219.0.0	Samsung Electronics Co., Ltd.
169.018.0.0	J.P. Morgan & Co.

Classe of IP adresses

Classe C	1	1	0	ID.RÉSEAU	ID.MACHINE

- C-Class :for little companies.
- **192.0.0.1** to **223.255.255.254**
- 24 bits **Net_ID** (2097152 networks)
- 8 bits for **Host_ID** (2^8-2 =254 Machines/network)

Classe of IP adresses

- D-Classe :Réserved for multicast.
- 224.0.0.0 à 239.255.255.255
- Example diffusion of routing tables.
- No need to extract Host_ID from IP, because DATA are not for a particular host.

Classe of IP adresses

- Classe E :Reseved for experimentations
- 240.0.0.0 to 255.255.255.255

Classe of IP adresses

- L'adresse d'un réseau est définie par un **Host_ID** égale à **0**.
- Exemple:
- L'IP 37.200.19.22 appartient au réseau 37.0.0.0 (classe *A*).
- L'IP 137.194.120.12 appartient au réseau 137.194.0.0 (classe *B*).
- L'IP 197.194.192.36 appartient au réseau 197.194.192.0 (classe *C*).

Broadcast Adress

- The broadcast adress of a network is defined by a **Host_ID** bits to **1**.
- Broadcast data are sent to all machines in the network.
- The broadcasst of network **37.0.0.0/8** is **37.255.255.255** (classe *A*).
- The broadcasst of network **137.194.0.0/16** is **37.194.255.255** (classe *B*).

Addresses

- **0.0.0.0** : used by a machine that has not an IP address.
- **255.255.255.255**: broadcast adress.

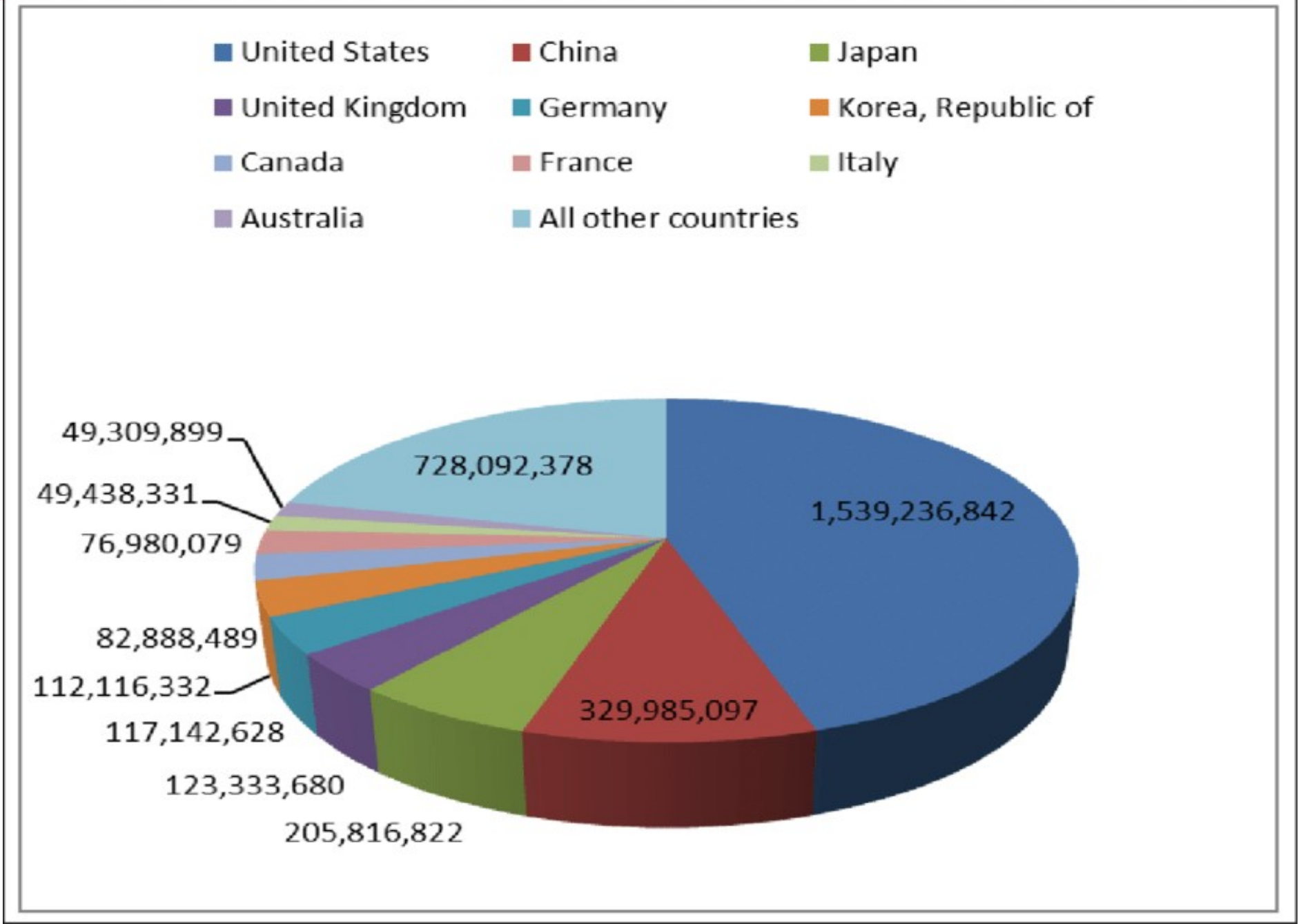
Classe d’adresses IP

- A =50%
- B=25%
- C=12.5%
- D=6.25%
- E=6.25%
- A bad distribution of IP adresses.
- Waste of IP addresses.

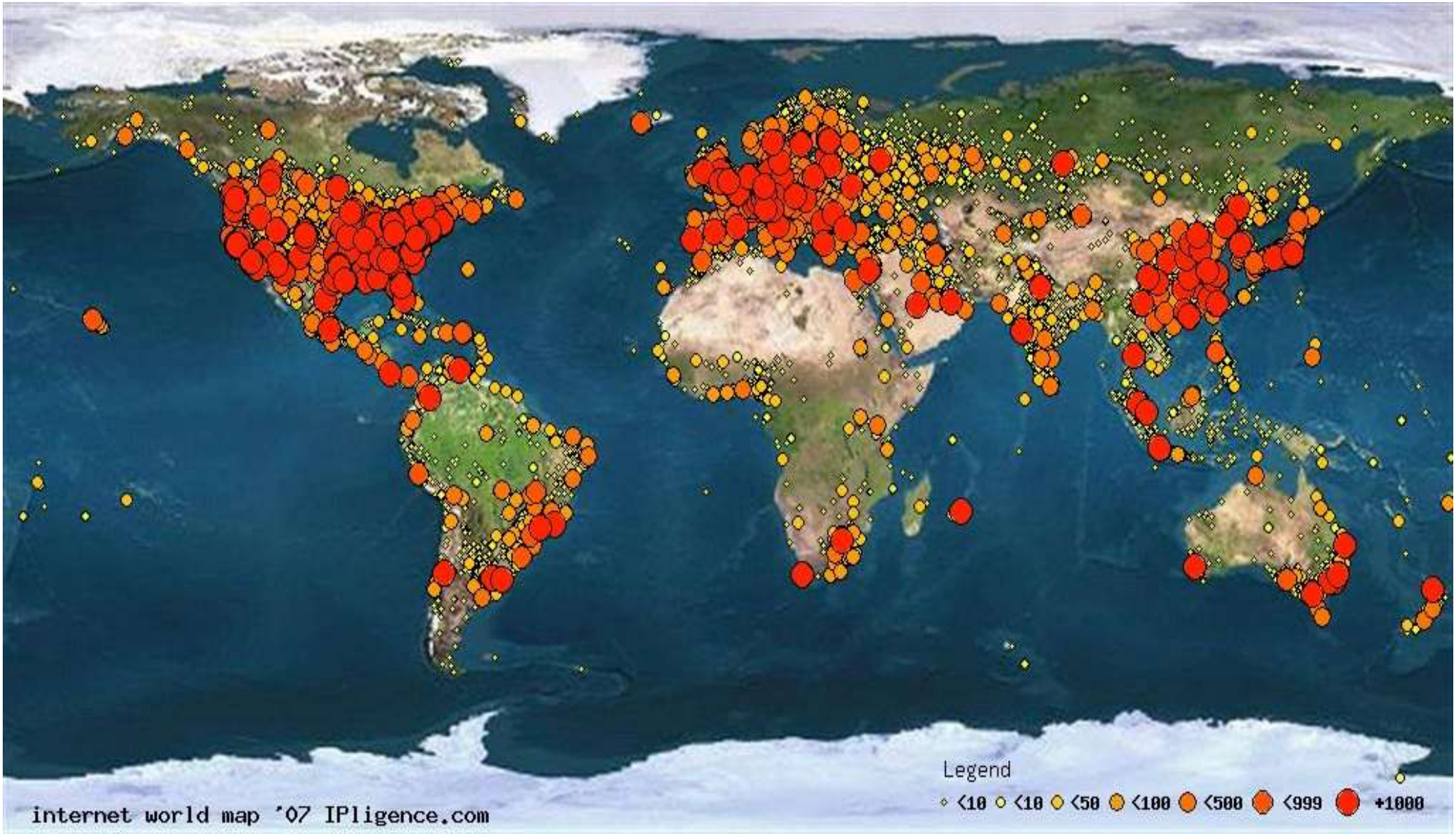
Classe d’adresses IP

Class	% of IP addresses
A	50%
B	25%
C	12,5%
D	6,25
E	6,25%

IP Distribution by countries



IP Distribution by countries



Public and Private Addresses

- IP addresses must be unique.
- There is no enough public IP Addresses. **(shortage of IP addresses)** ???
- Majority of networks doesn't require external interconnection (private network), for security reasons.
- IP address uniqueness is not mandatory.

Public and Privées Addresses

- L'IANA Has reseved address ranges to use in private networks.
- Addresses asigned internally to device on private network.
- RFC 1918. "**Address Allocation for Private Internets**"
- These adresses are **non-routables on** internet network.
- Hosts with these addresses are only visible in private network.

Private Networks Addresses

Each Class has its own private address range.

class	Start	End	Nunberof network
Classe A	10.0.0.0	10.255.255.255	1
Classe B	172.16.0.0	172.31.255.255	16
Classe C	192.168.0.0	192.168.255.255	256

Sub-network

- Since IPs are rare.
- To acquire them you have to pay.

- Suppose **Université Constantine 2** need a IP range of 4 public IP.
- Same for **Université Constantine 3**

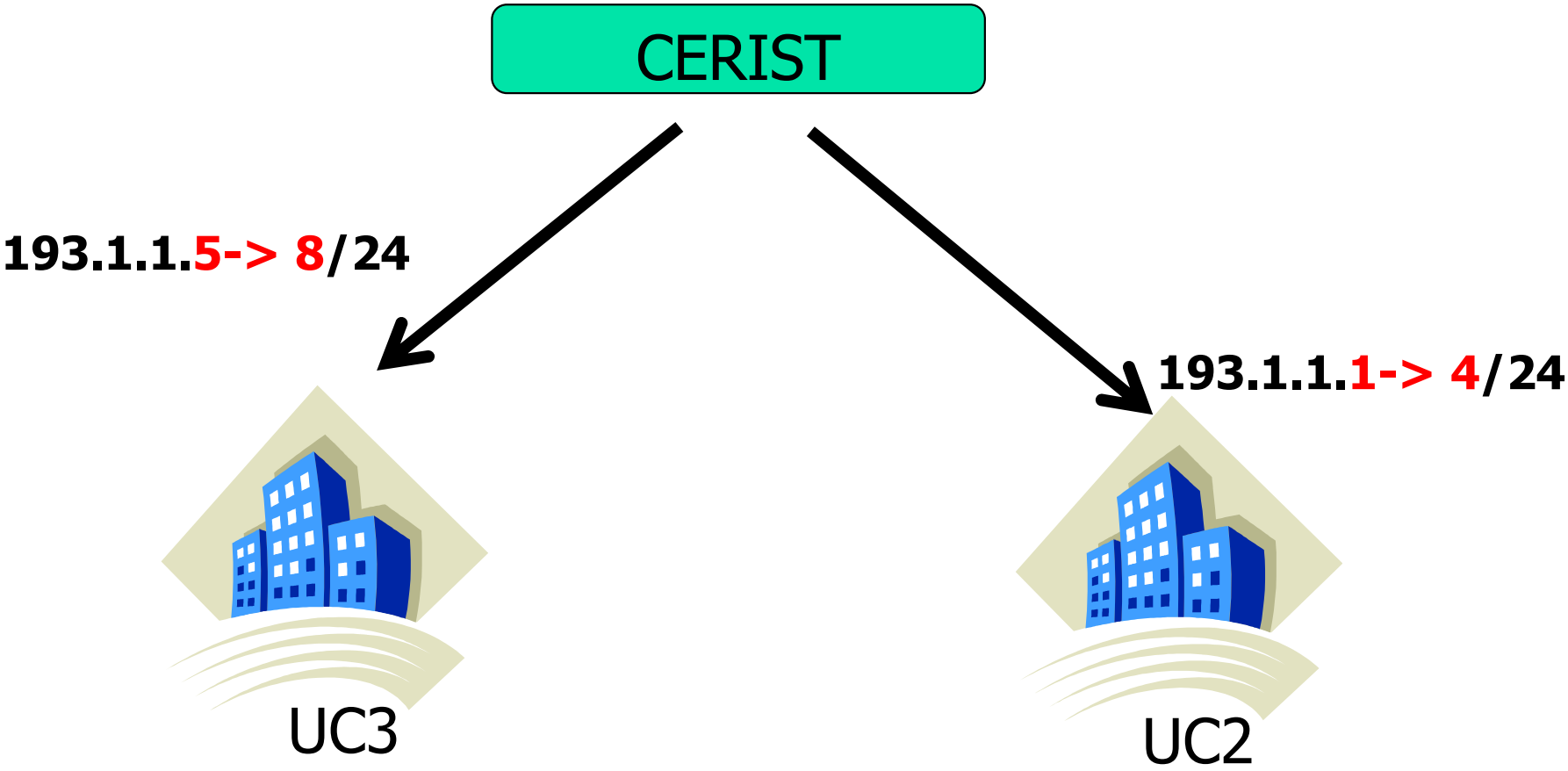
Sub-Net

- Suppose Internet service providers **CERIST**) propose a C-address.
- 193.1.1.0 with mask 24.
- Used IP utilisées are :
- 193.1.1.**1** to 193.1.1.**4**
- Not used addresses :
- 193.1.1.**5** to 193.1.1.**254** (250 IP).

Sub-Net

- Assigned Addresses 193.1.1.**5** to 193.1.1.**8**
- Université Constantine 3.

Sub-Net



Routing problem at CERIST for the net **193.1.1.0**

Same Network 193.1.1.0/24

Sub-Net

- How to use the remaining addresses ????
- Change the network mask to get smaller networks.
- 193.1.1.0 255.255.255.0

1st Address	**193.1.1.1**
Last Address	**193.1.1.254**
Broadcast	**193.1.1.255**

Sub-Net

193.1.1.0	193.1.1.00000000	Net address
193.1.1.1	193.1.1.00000001	@ 1st Machine
193.1.1.127	193.1.1.011111111	
193.1.1.128	193.1.1.10000000	
193.1.1.254	193.1.1.11111110	Last Machine
193.1.1.255	193.1.1.11111111	Broadcast

We will change the network mask to (25 bits)

255.255.255.10000000

Sub-Net

We'll create two subnetworks

193.1.1.0	193.1.1.00000000	1st net	1st subnet 193.1.1.0/25 126 Machines
193.1.1.1	193.1.1.00000001	@ 1st machine	
193.1.1.126	193.1.1.01111110	@ Last machine	
193.1.1.127	193.1.1.01111111	@ Broadcast 1st net	

193.1.1.128	193.1.1.10000000	@ du 2eme réseau	2nd subnet 193.1.1.128/25 126 Machines
193.1.1.129	193.1.1.10000001	@ 1st machine	
193.1.1.254	193.1.1.11111110	@ last machine	
193.1.1.255	193.1.1.11111111	@ Broadcast. 2nd net	

N=2 **255.255.255.11000000=255.255.255.192**

193.1.1.00000000
193.1.1.00000001

193.1.1.00111110
193.1.1.00111111

1st Subnet
193.1.1.0/26

193.1.1.01000000
193.1.1.01000001

193.1.1.01111110
193.1.1.01111111

2nd Subnet
193.1.1.64/26

193.1.1.10000000
193.1.1.10000001

193.1.1.10111110
193.1.1.10111111

3th Subnet
193.1.1.128/26

193.1.1.11000000
193.1.1.11000001

193.1.1.11111110
193.1.1.11111111

4th Subnet
193.1.1.192/26

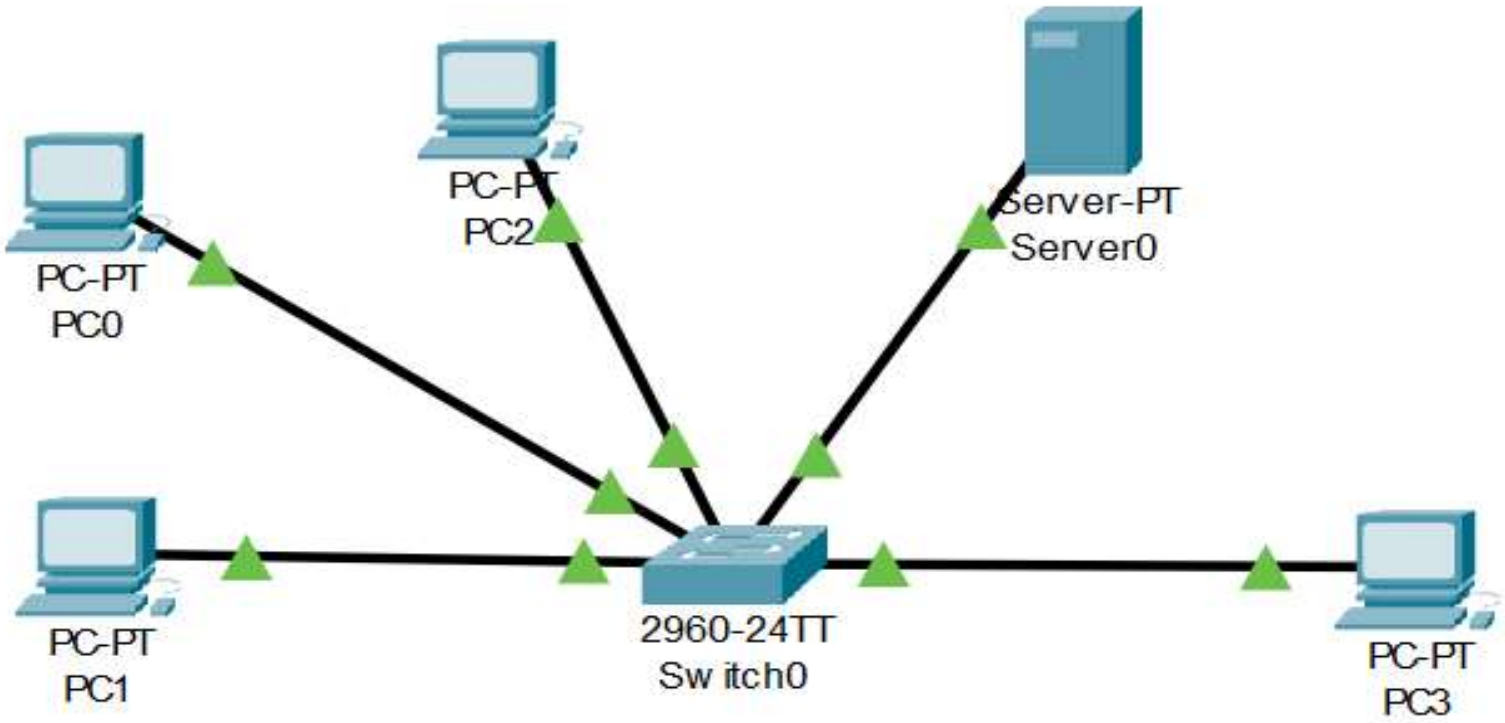
4 Subnets

62 Machines per Subnet

Sub-Net

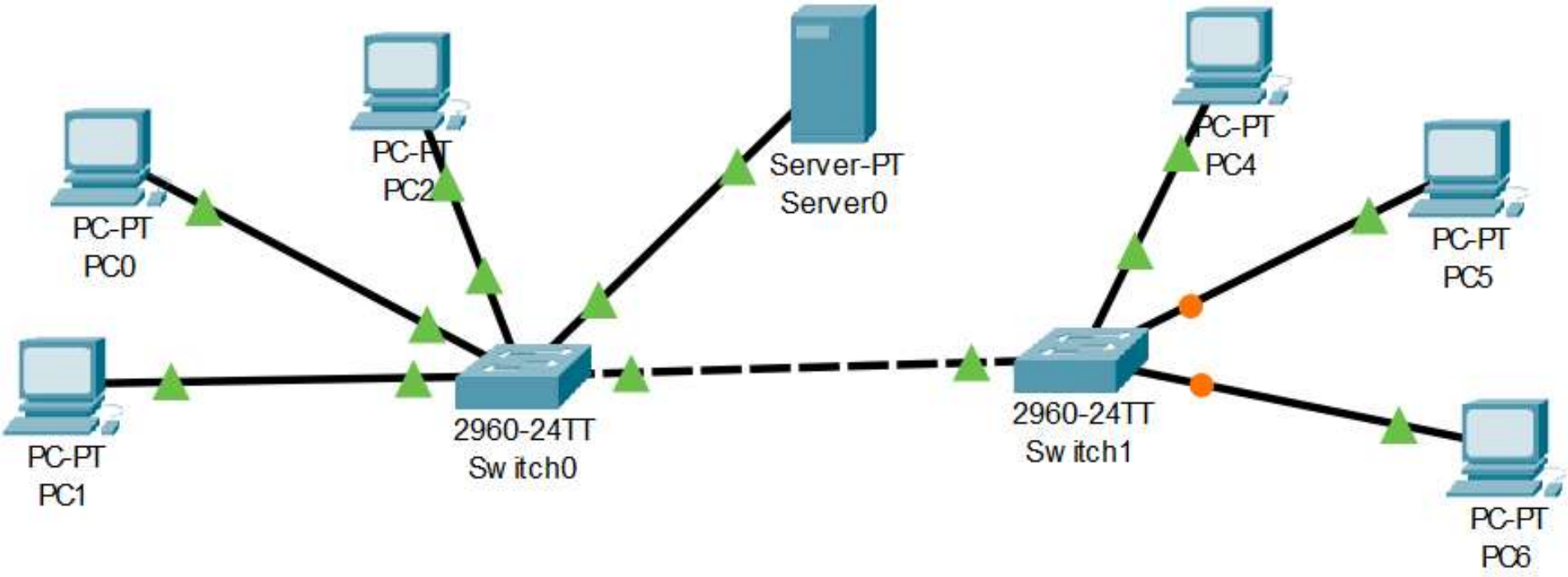
n	Mask	Nbr Subnet	Machines per Subnet
1	255.255.255.128	2	126
2	255.255.255.192	4	62
3	255.255.255.224	8	30
4	255.255.255.240	16	14
5	255.255.255.248	32	6
6	255.255.255.252	64	2

Routing : broadcast domain

- A **broadcast domain** is a logical division of a computer network, in which all nodes can reach each other by broadcast at the data link layer.
- 

Routing : broadcast domain

- A broadcast can be on cascading switches.
- 192.168.20.0/24



Routing : Switch

- Switches connect devices in a network and use packet switching to send, receive or forward data packets or data frames over the network.
- They are networking devices operating at layer 2 or a **data link layer** of the OSI model.
- They user MAC addresses to communicate.

Routing : Switch

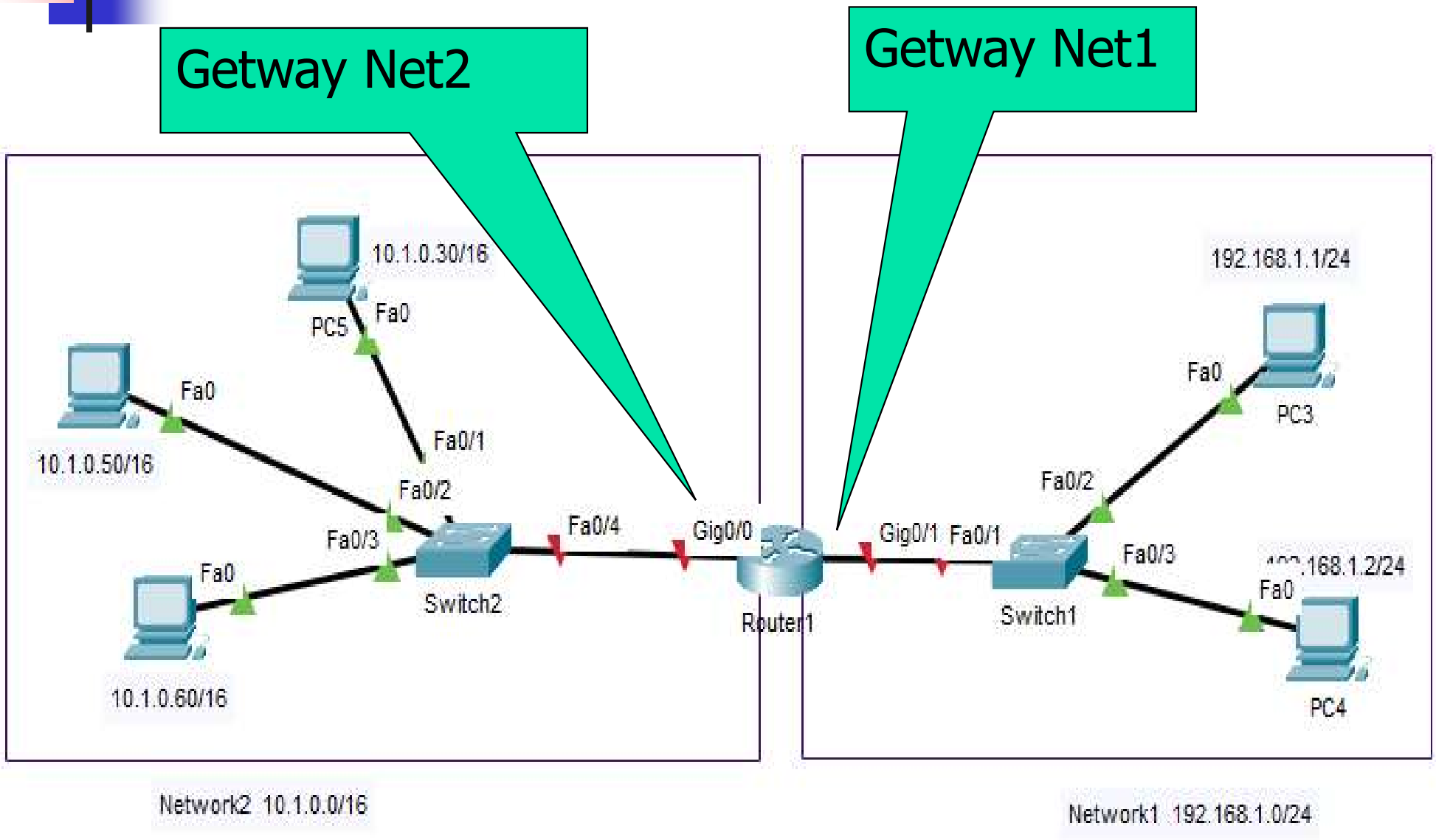
- A Switch manage a table that indicates for each port the connected MAC address of equipement.
- show mac-address-table

- **Switch#show mac-address-table**
- Mac Address Table
- ----------
- Vlan **Mac Address** Type **Ports**
- ---- ---------- -------- ----- ---------- ----------
- 1 **0001.63ce.b2e3** DYNAMIC **Fa0/4**
- 1 **0001.c935.18d4** DYNAMIC **Fa0/3**
- 1 **000d.bd54.693c** DYNAMIC **Fa0/2**
- 1 **00d0.d3c7.5473** DYNAMIC **Fa0/1**
- Switch#

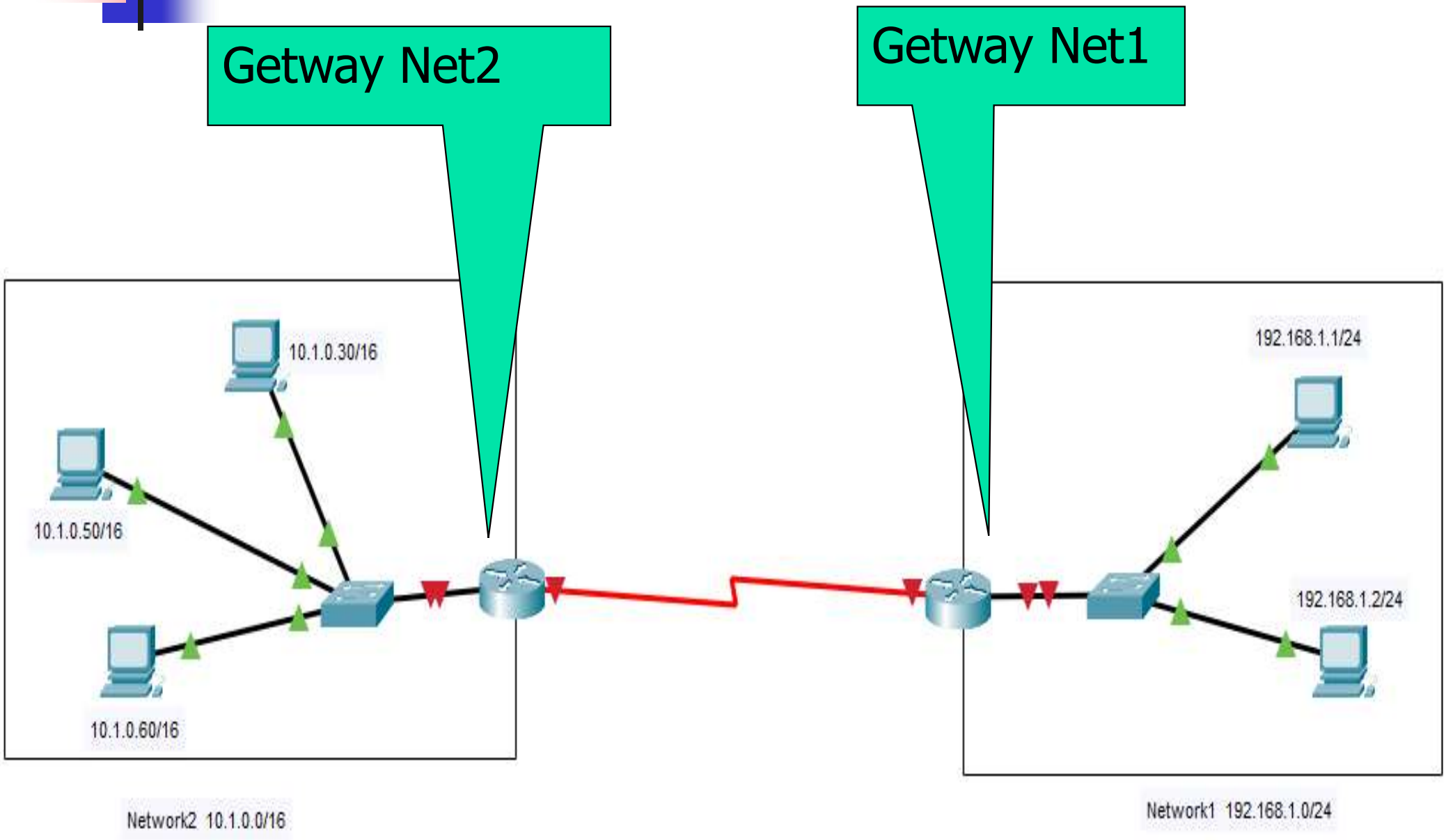
Router

- A device that forwards data from one network to another. It interconnects two or more IP network.
- Router operate on layer 3 (IP).
- Based on an internal routing table and the address of the destination network in the incoming packet.
- A router contains at least two network cards.

Router


Getway Net2
Getway Net1
10.1.0.30/16
PC5
Fa0
Fa0
10.1.0.50/16
Fa0/1
Fa0/2
Fa0/3
Fa0
10.1.0.60/16
Switch2
Fa0/4
Gig0/0
Router1
Gig0/1
Fa0/1
192.168.1.1/24
Fa0
PC3
Fa0/2
Switch1
Fa0/3
Fa0
PC4
Network2 10.1.0.0/16
Network1 192.168.1.0/24

Router

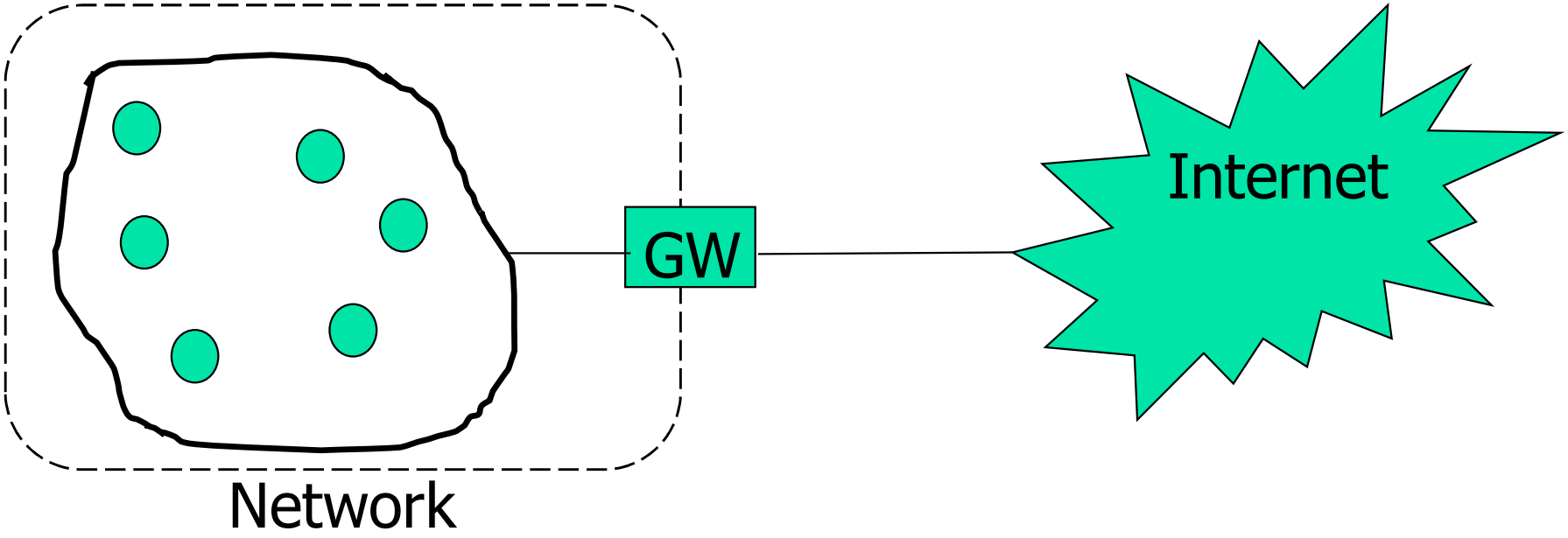

Getway Net2
Getway Net1
10.1.0.30/16
10.1.0.50/16
10.1.0.60/16
192.168.1.1/24
192.168.1.2/24
Network2 10.1.0.0/16
Network1 192.168.1.0/24

Gateway

- A gateway is a single node (computer or router) on a network that serves as an entrance to another network.
- For enterprises, a gateway is an individual computer that is responsible for routing traffic from a workstation to an outside network.
- A workstation that has to sent a data to an outside network it gives the data to the gateway.

Gateway

- A gateway must have 2 interfaces



config t
config t
config t

Router IP configuration

- Firstly, enter the command line by clicking on any router and then selecting the **CLI** tab.
- Enter **config mode** by typing the following commands in the command line:
  - en
  - conf t

Router>en

Router>enable

Router#conf t <- configure terminal mode

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#

config t
config t
co

FastEthernet

- Select the interface by typing **interface** followed by the label of the interface
  - (fa->FastEthernet, g->GigabitEthernet).

- For example int g0/1 -→ interface GigabitEthernet 0/1

Router(config)#int g0/1

Router(config-if)#

Interface mode

- enter the ip with net-mask by command ip add

Router(**config-if**)# ip address 192.168.1.1 255.255.255.0

Router(**config-if**)# no shutdown ←-to keep interface **on**

Router(**config-if**)# exit ←---to return conf mode

Routing

- Etablissent des règles d'échange entre routeurs pour mettre à jour leurs tables selon des critères de coût comme:

  La distance, l'état de la liaison, le débit.
- Afin d'éviter les problèmes de boucles de routage et gestion des pannes dans le réseau.

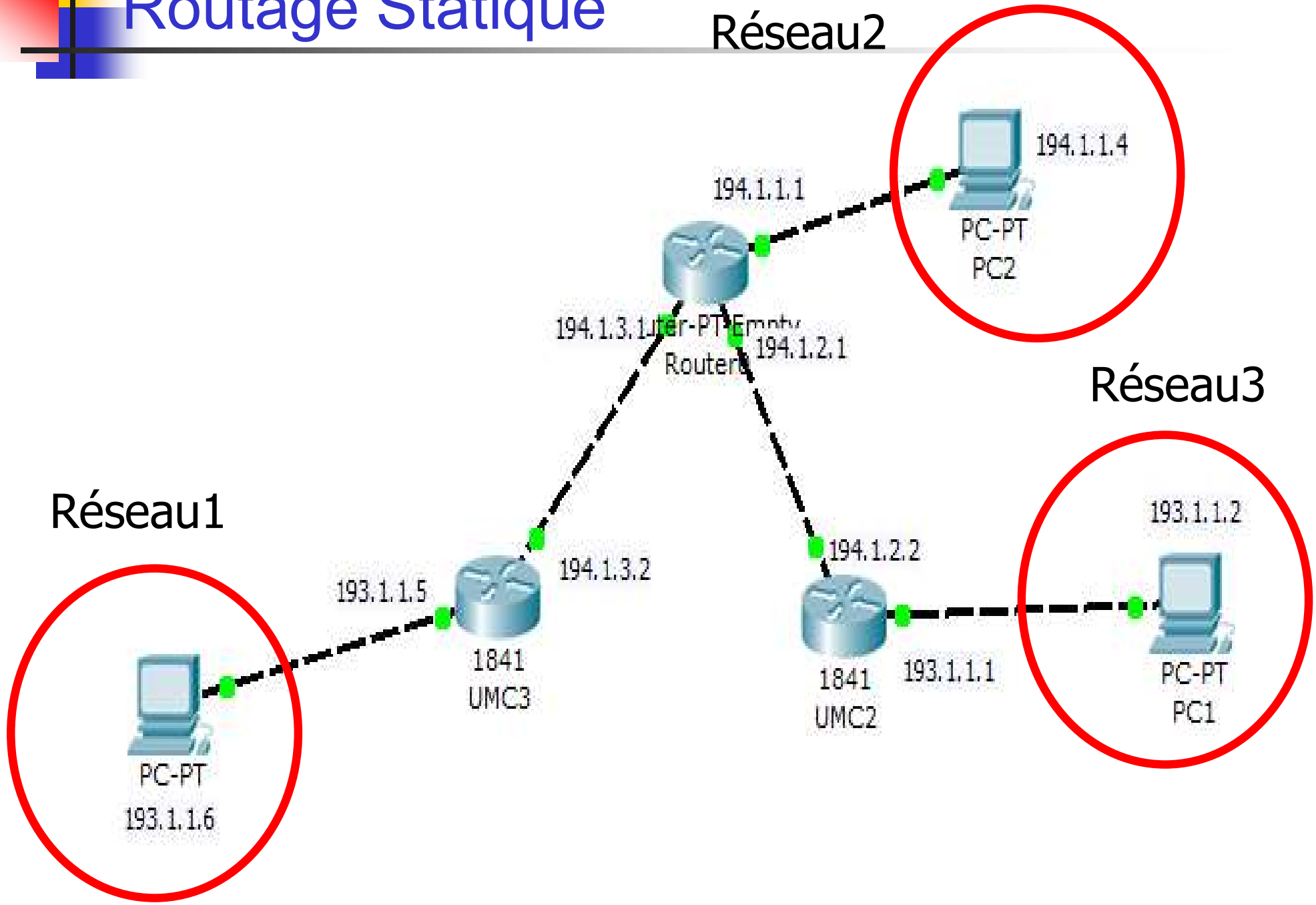
Static Routing

- Every host on the Local network needs to know which router to exit through to reach an external network.
- Routing informations are stored in routing tables.
- Routing tables contain the IPs of the routers for the different paths to be taken.

Show ip route

- *To display routing table*

Routage Statique



Static Routing

- With static routing, routing tables are updated manually whenever the network topology changes.
- To add a route on the router, use **ip route**

router(config)#ip route network_ip interface

- Interface : the interface which leads to the network
- *ip route 192.168.1.0 255.255.255.0 g0/0*

Static Routing

- Show ip route
-

- 10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
- C 10.1.1.0/24 is directly connected, Serial0/3/0
- L 10.1.1.2/32 is directly connected, Serial0/3/0
- **S 192.168.1.0/24 is directly connected, Serial0/3/0**
- 192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
- C 192.168.3.0/24 is directly connected, GigabitEthernet0/0
- L 192.168.3.1/32 is directly connected, GigabitEthernet0/0
-

  Router#

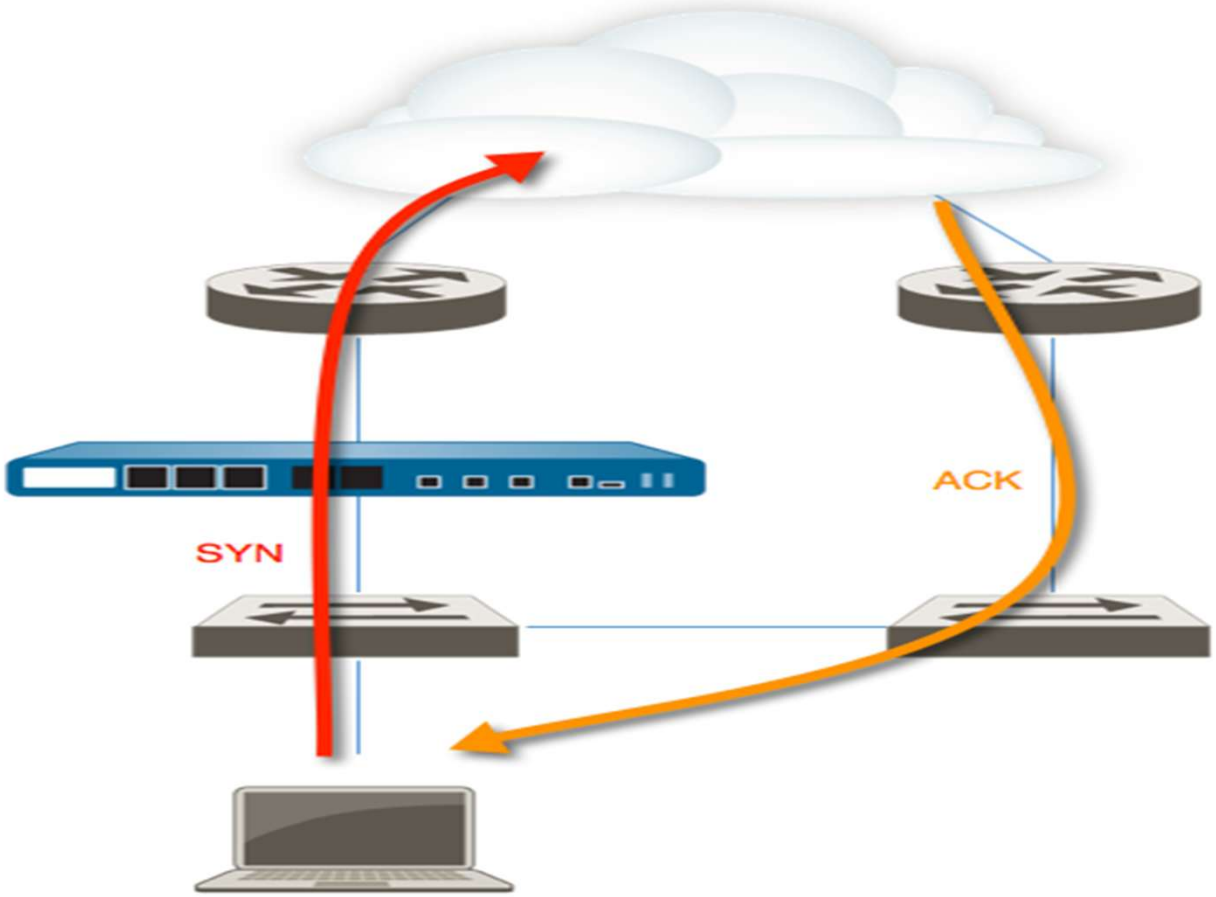
Static Routing

- **avantage**
- Security of certain parts of the network through masking.
- Less overhead than dynamic routing.
- No change of routing information.

Static Routing

- **Inconvénients:**
- Manual routing management.
- A station can only reach the networks indicated by the route table.
- The risk of looping.
- Asymmetric routing

Static Routing : Asymmetric routing



Dynamic Routing

- Routing protocols define the rules for the dynamic exchange of information between routers to build the routing table for a routed protocol.
- Build dynamically the path between routers.
- The routed protocol will then use that information to drive the packets and find the best path to a destination.

Dynamic Routing

- **RIP** (*Routing Information Protocol*).
- **OSPF** (Open Shortest Path First).
- **EIGRP** ((Enhanced Interior Gateway Routing Protocol))

Dynamic Routing

- Routing protocols define the rules for the dynamic exchange of information between routers to build the routing table for a routed protocol.
- Build dynamically the path between routers.
- The routed protocol will then use that information to drive the packets and find the best path to a destination.

Dynamic Routing

- Depends on the metric:
- Metric: a cost assigned for passing through a network.
- A router should choose the route with the smallest metric.
- The metric assigned to each network depends on the type of protocol.

Dynamic Routing: RIP

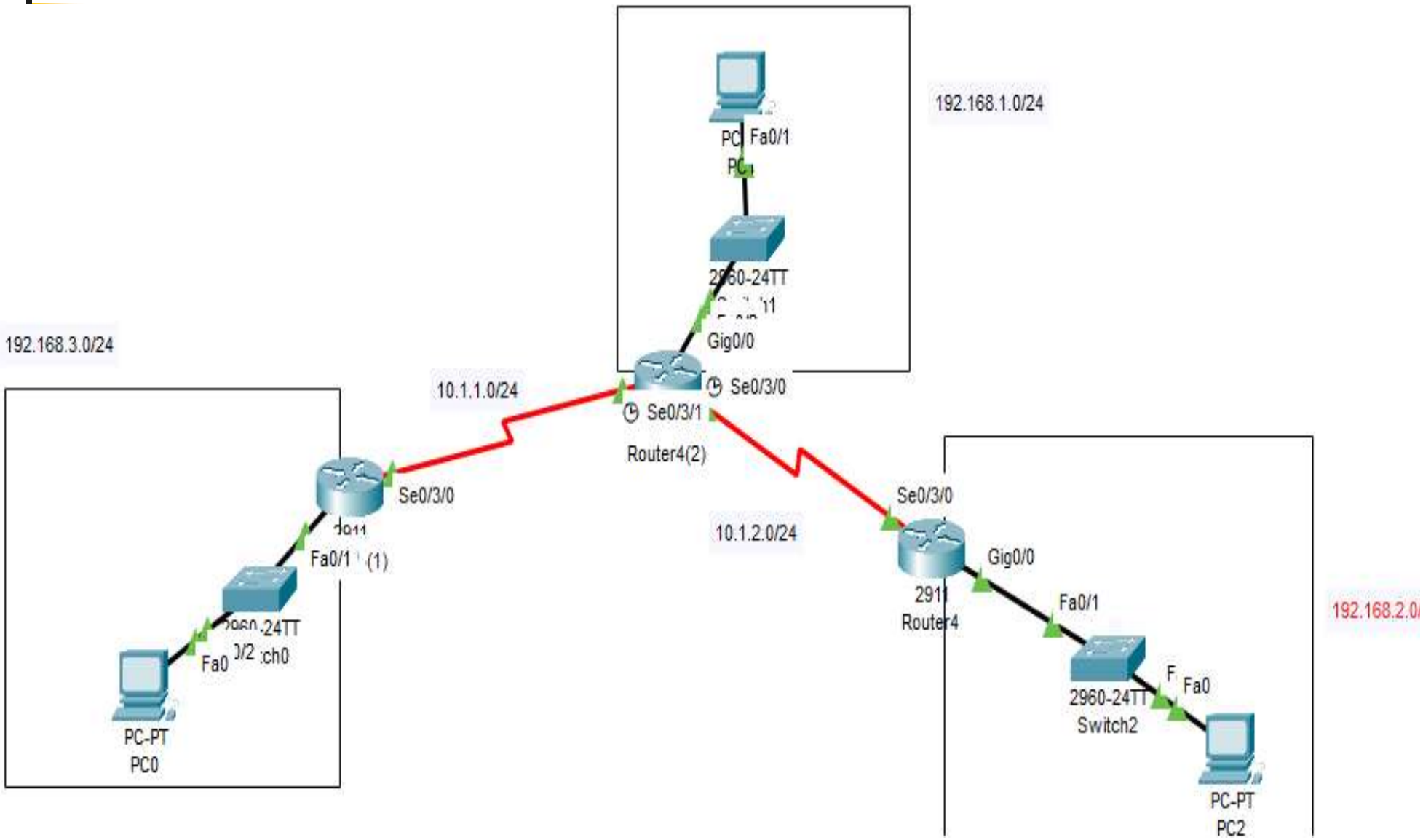
- Based on distance vector routing.
- Use the **Bellman-Ford** algorithm for calculating the routing tables.

Dynamic Routing: RIP

- To configure a routing RIP on a router you have to select RIP protocol and gives the networks connected directelly to the router.
- *Router(config)# router rip*
- Than gives networks connected to router
- *Router(config-router)# network net*

Example

- *Router(config-router)# network* ***192.168.1.0***


192.168.1.0/24
Fa0/1
Gig0/0
Se0/3/0
Se0/3/1
Router4(2)
10.1.1.0/24
192.168.3.0/24
Se0/3/0
Fa0/1
(1)
Fa0
PC-PT
PC0
10.1.2.0/24
Se0/3/0
Gig0/0
Fa0/1
Router4
Fa0
2960-24TT
Switch2
PC-PT
PC2

Configuration Réseaux sous Centos

- Les interfaces réseaux sous Linux sont les composants qui permettent au système de communiquer avec d’autres ordinateurs.

- Ces interfaces peuvent être:
- Ethernet :réseaux câblés.
- Wifi : réseaux sans fils.

Configuration Réseaux sous Centos

- La liste des interfaces réseaux peut être affiché par les commandes:
- **`ip  link show`**
- **`ip  a`**
- **`ifconfig ancienne commande`**
- Le fichier **/proc/net/dev** :donne le trafic effectué par les différents interfaces réseaux.

Configuration Réseaux sous Centos

- Les interfaces Ethernet sont appelées: **eth0, eth1** …
- Les interfaces Wifi sont nommées : **wlan0, wlan1**, …
- Les interfaces PPP
- **PPP0, PPP1,…**
- **lo**: interface de bouclage(loopBack): utilisé pour les communication entre applications de la même machine.

Configuration Réseaux sous Centos

- L’attribution d’adresses IP aux interfaces réseau s’effectue d’une façon:
- **Dynamique:** l’adresses est attribuée par un serveur DHCP
- **Statique**: l’adresse est introduite manuellement.

Configuration Réseaux sous Centos

- Le répertoire :
- **`/etc/sysconfig/network-scripts/`**
- Comporte les fichiers des interfaces réseau:
- **`ifcfg-enp03`** : L'interface **eth0**
- **`ifcfg-enp08`** : L'interface **eth1**
- **.....**

Configuration Réseaux sous Centos

/etc/sysconfig/network-scripts/ifcfg-eth0

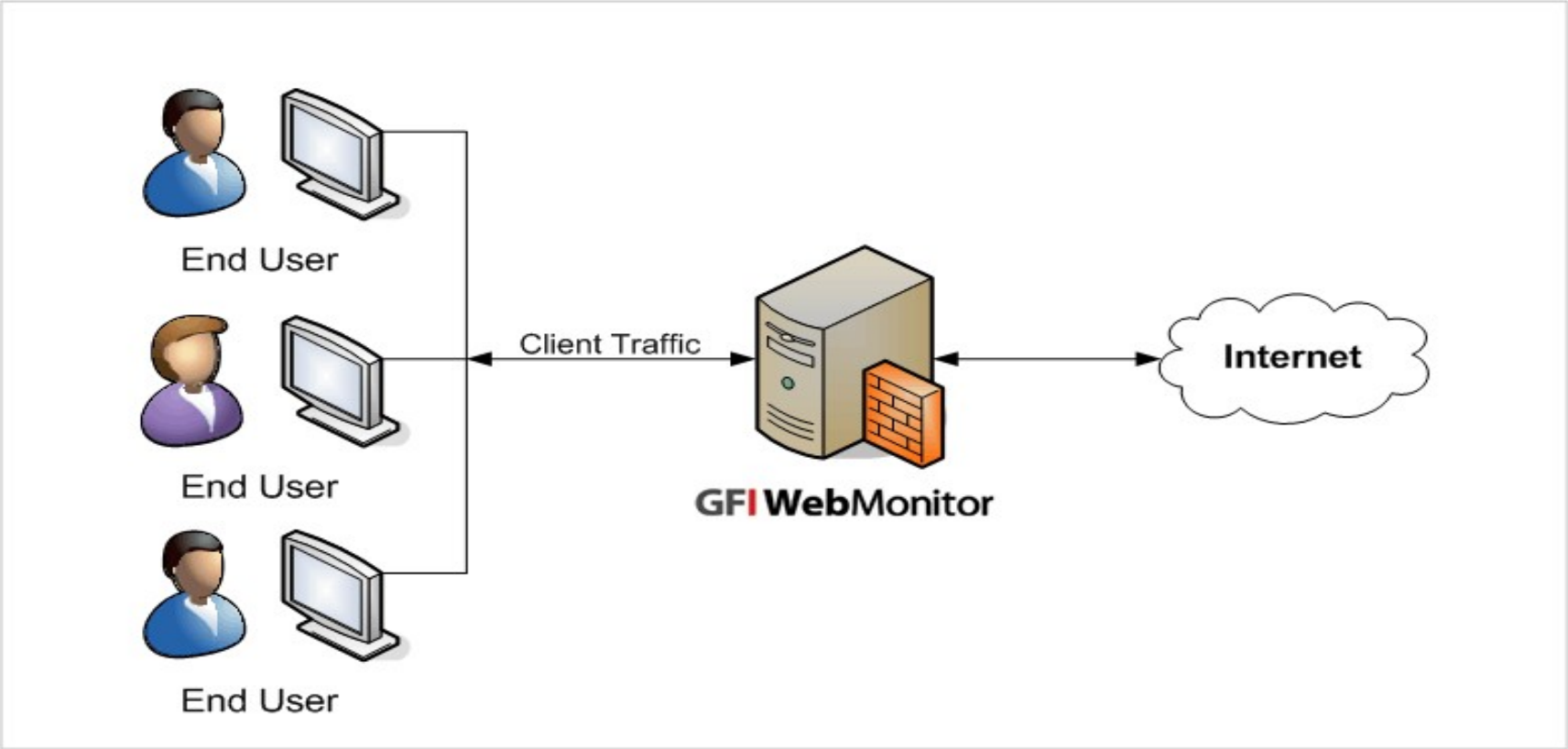
```
DEVICE=eth0
BOOTPROTO=static
HWADDR=XX :XX :XX:XX:XX:XX
ONBOOT=yes
TYPE=Ethernet
UUID="generated by the os"
IPADDR=192.168.1.252
NETMASK=255.255.255.0
```

Configuration Interface réseau

- **DEVICE** : Nom de l’interface.
- **ONBOOT** : activation de l'interface au démarage.
- **BOOTPROTO** : protocole (***DHCP*** ou ***static***)
- **NETMASK** : masque de sous-réseau.
- **IPADDR** : adresse IP du réseau

Configuration Interface Réseau

- **Passerelle** (gateway): est un système matériel et logiciel permettant de faire la liaison entre deux réseaux.



Les outils d'administration

- le fichier **/etc/sysconfig/network**

```
NETWORKING=yes
HOSTNAME=server.ec0.local
GATEWAY=192.168.1.254
```

Les outils d'administration

- Ensemble de fichiers qui permettent de configurer le réseau d'une machine.
- **`/etc/hosts`,**
- **`/etc/networks`,**
- **`/etc/host.conf`,**
- **`/etc/resolv.conf`**

/etc/hosts

- Permet de définir les **FQDN** (Fully Qualified Domain Name).
- Le *Nom d'hôte pleinement nommés* des hôtes.
- Permet le mapping des adresses IP en des nom d’hôtes (sans recourir à un DNS).

etc/hosts

```
#nom local pour le bouclage
127.0.0.1 localhost localhost.localdomain
#autres hotes.
192.168.1.1 smail.uc2.net      smail
192.168.1.2 ssquid.uc2.net     ssquid
192.168.1.3 sldap.uc2.net      sldap
```

@ IP

FQDN

Alias

/etc/networks

- Permet de donner des **noms logiques** à des réseaux, chaque entrée doit se terminer par un 0.
- Exemple

```
default 0.0.0.0
loopback 127.0.0.0
link-local 169.254.0.0
```

/etc/host.conf

- Donne l'ordre dans lequel le processus de résolution de noms est effectué.
- Le mapping des adresses IP en des FQDN.
- L’option ***order*** définie l’ordre de résolution.
- L’option ***multi on/off*** spécifie que les machines peuvent avoir plusieurs adresses ip.

/etc/host.conf

```
#Lookup names via DNS first then fall back
  to /etc/hosts.
order bind,hosts
#We have machines with multiple IP addresses
   multi on
```

/etc/resolv.conf

- Permet de définir la liste des serveurs de noms (DNS) à utilisés.
- Exemple:

```
nameserver 8.8.8.8
nameserver 8.8.4.4
```

ip

- **`ip:`** permet la configuration locale ou à distance des interfaces réseau.
- ip a (ip addr show) : affiche la liste des interfaces réseaux.
- **`ip addr add`** ***`ip_adr`*** **`dev`** ***`net_dev`***
- **`ip addr add 192.168.2.221/24 dev eth0`**
- **`Attribut l'adresse ip à l'interface eth0. (dans la configuration dhcp de l'interface)`**

ip

- ip addr del *ip_adr* dev *net_dev*
- ip addr del 192.168.2.221/24 dev eth0
- Suprime l'adresse ip à l'interface eth0. (dans la configuration dhcp de l'interface)

arp (Adress Resolution Protocol)

- Le protocole ***arp*** effectue la conversion des adresses **IP** en des adresses **MAC**.
- Lorsque un ordinateur envoi des données vers un autre ordinateur dont il connait l'adresse IP et ne connait pas l'adresse physique (MAC).
- l'ordinateur émetteur émet une requête ***arp*** en **broadcast**.

arp (Adress Resolution Protocol)

- L'ordinateur qui a l'***IP***, ajouteras l'***IP*** et **MAC** de l'émetteur dans sa table **arp** et
- Répondras par une requête ***arp*** « **je suis *adresseIP*, mon adresse MAC est *adresseMAC*** ».
- A la réception de la réponse, l'émetteur ajouteras l'IP et MAC du destinataire dans sa table ***arp***.

route, netstat, traceroute, dig, host

- **`route`** : affiche et manipule la table de routage.
- **`route -n`**
- **`route add default gw 192.168.1.10`**
- **`route -Cn`** : Kernel's Routing Cache Information.
- **`route add -host 192.168.1.51 reject`**

Reject Routing to a Particular Host or Network

route, netstat, traceroute, dig, host

- **`netstat :`** affiche diverses informations sur le réseau telles que les connexions, tables de routage , stat des interfaces…..
- **`-a`** : Tous les ports
- **`-t`** : Tous les ports TCP
- **`-u`** : Tous les ports UDP
- **`-l`** : Tous les ports en écoute.
- **`-n`** : Affiche les IP. Pas de résolution de nom.
- **`-p`** : Affiche le nom du programme et le PID associé.

route, netstat, traceroute, dig, host

- **`dig`** : un outils permettant d'interroger des serveurs DNS.
- **`dig @DNS-server-name IPAddress`**
- **`dig @8.8.8.8 sun.com`**

- Par défaut la commande est non installé.

route, netstat, traceroute, dig, host

- **`host :`** un outils DNS qui converti un nom en adresse IP ou inversement.
- **`host @nom`**
- **`host ip_adr`**

- Par défaut la commande est non installé.

Firewalld

- Firewall zone.
- Zones are set of rules that allow a specific traffic and network services based on various trust of levels.
- Each zone allows different type of services and ports.
- When enabled FirewallD for the first time the “**public**” will be default zone.

Firewalld

- **Drop Zone** - This is the lowest level of trust. Only outgoing connections are allowed in this zone. All incoming connection are dropped without any reply.
- **Block zone** - Block zone is similar to drop zone. The only difference is, if we use this incoming request is rejected with icmp-host-prohibited message.
- **Public zone** - Selected connections are accepted by defining rules in this zone other connections will be dropped.

Firewalld

- **External zone** - This zone is for external networks with masquerading is enabled other connections will be rejected only specified connection will be allowed.
- **DMZ zone** - Demilitarized zone provides limited access to your internal network. Only specified connections will be allowed.
- **Work Zone** - This zone is used for work areas. Most of the computers are trusted in the network. Only specified connections will be allowed.

Firewalld

- **External zone** - This zone is for external networks with masquerading is enabled other connections will be rejected only specified connection will be allowed.
- **Home zone** - used for home areas. Few more connections are allowed.
- **Internal zone** - used for internal networks with selected connections allowed.
- **Trusted zone** - Trust all the network connections. If we set this zone then all traffic will be accepted.

Firewalld

- Default zone

```
firewall-cmd --get-default-zone
```

- which zone is active

```
firewall-cmd --get-active-zone
```

- change the default zone

```
firewall-cmd --set-default-zone=internal
```

- list all active services, ports, rules

```
firewall-cmd --list-all
```

Firewalld

- Firewalld configuration
- Firewalld Services is configured with XML files located at /usr/lib/firewalld/services/

- [root@localhost services]# **cat http.xml**
- <?xml version="1.0" encoding="utf-8"?>
- <service>
- <short>**WWW (HTTP)**</short>
- <description>HTTP is the protocol used to serve Web pages. If you plan to make your Web server publicly available, enable this option. This option is not required for viewing pages locally or developing Web pages.</description>
- **<port protocol="tcp" port="80"/>**
- </service>
- [root@localhost services]#

Firewalld

- Add the rule **http**

```
firewall-cmd --zone=public --add-service=http
```

- Add the rule **http** permanent (it remain after reboot)

```
firewall-cmd --zone=public --add-service=http --permanent
```

- Opening a Port for a zones

```
firewall-cmd --permanent --zone=public --add-port=80/tcp
```

- open a range of ports (200 to 300)

```
firewall-cmd --permanent --add-port=200-300/tcp
```

- remove an added port
- `firewall-cmd --zone=public --remove-port=5000/tcp`

THANK YOU